Industrial Chemistry

Topic 14

Key ideas

Unit 45 An introduction to industrial chemistry
- Importance of chemical processes in industry
- Rate equation
- Order of reaction

Unit 46 Factors affecting the rate of a reaction
- Activation energy
- Maxwell-Boltzmann distribution curve
- Arrhenius equation
- Characteristics of catalysts

Unit 47 Industrial processes
- Conversion of ammonia to fertilizers
- Chloroalkali industry
- Conversion of methane to methanol

Unit 48 Green chemistry
- Sustainable development
- Principles of green chemistry
- Atom economy
- Using the manufacture of acetic acid as an example to show the practice of green chemistry principles in industry









Unit 45 An introduction to industrial chemistry

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

45.1 What is chemical industry?

In our modern industrial society there is a great demand for synthetic products that are made from raw materials found in the Earth's crust, ocean and atmosphere. Such products include paint, plastics, fertilizers, steel, glass, paper, cement and *pharmaceuticals*.

The chemical industry converts raw materials found in the Earth's crust, ocean and atmosphere into products of greater value to us.

The basic raw materials of the chemical industry include:

- petroleum and coal;
- air;
- water;
- vegetable materials; and
- minerals such as metal ores, common salt, limestone and *gypsum*.



Fig. 45.1 The chemical industry converts raw materials into products of greater value to us

pharmaceuticals 藥品　　gypsum 石膏

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

For each of the following products, suggest the raw material that is used to make it.



1 Plastic shopping bag
Petroleum



2 Household chlorine bleach Sea water



3 Soft drink can
Bauxite



4 Car battery acid
Sulphur

45.2 Products from the chemical industry

The chemical industry produces five categories of product ranging from bulk chemicals to high-value chemicals produced in small amounts:

- basic inorganic chemicals and fertilizers — the industry makes large amounts of basic inorganic chemicals such as sulphuric acid and nitric acid, sodium hydroxide, chlorine and ammonia;
- petrochemicals and polymers — petrochemical plants use hydrocarbons from petroleum to make a great variety of products, including synthetic polymers, i.e. plastics*;
- dyestuffs, paints and pigments — modern dyes are made from petrochemicals;
- pharmaceuticals — the industry produces drugs and medicines;
- specialities — speciality chemicals are intermediates and end products required for the production of a wide range of goods, including food flavourings, liquid crystals, printing inks and agricultural chemicals, as well as formulations for personal care and cleaning.

* We have discussed the manufacture of plastics from chemicals derived from petroleum in Topic 7 Fossil Fuels and Carbon Compounds.



(a) Plastics



(b) Paints



(c) Drugs

Fig. 45.2 Plastics, paints and drugs are products of the chemical industry

N1 China's chemical industry

The chemical industry in China is maturing and will grow at a much more settled pace than during the previous boom years. The new but stable GDP growth rate of around 7%, along with rising domestic consumption, greater demand for auto and electronic products and ambitious sustainability targets will provide ample growth opportunities. China's chemical industry is estimated to grow between 9 and 11% in the period 2013 to 2015.

However, despite this growth, China has a net chemical deficit with the world market and is heavily dependent on imported materials. This dependency has been affected by price trends in the world market caused by heavy international demand for raw materials, petroleum and other chemical inputs.

Reference:

KPMG Advisory (China) Limited(2013) 'China's Chemical Industry: The Emergence of Local Champions', retrieved from http://www.kpmg.com/CN/en/IssuesAndInsights/ArticlesPublications/Documents/China-Chemical-Industry-201310.pdf (accessed May 2015)

Fig. 45.3 shows a geographic breakdown of world chemicals sales in 2013. The chemical industry is fundamental to many other industries. It often produces the basic material building blocks and intermediates for many other types of production and services. For example, the automobile industry uses enormous supplies of industrial chemicals such as paints and plastics for bumpers, dashboard panels and carpeting.

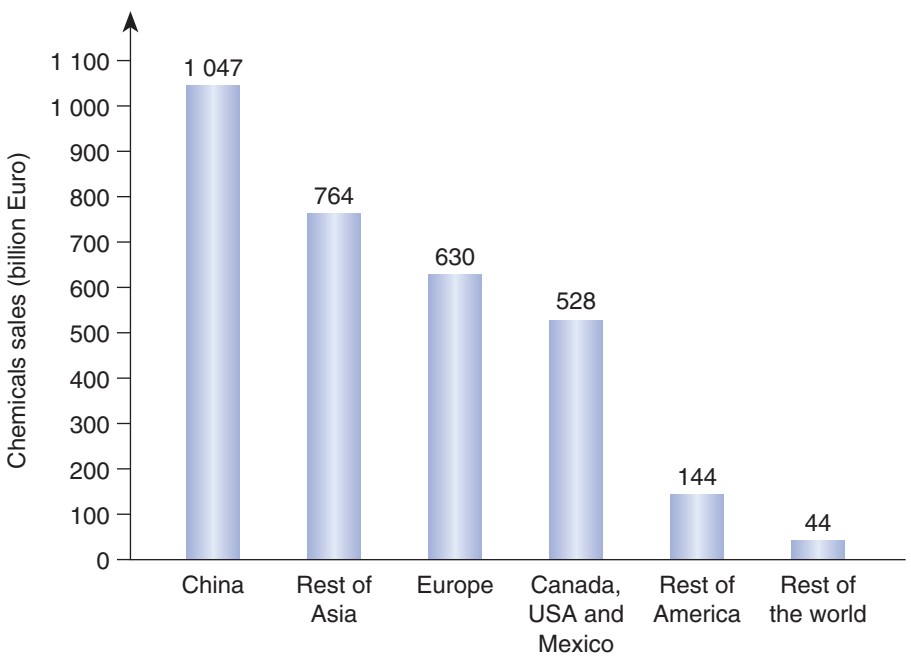


Fig. 45.3 Geographic breakdown of world chemicals sales in 2013 (Source: Cefic Chemdata International)

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Refer to page T15.

To understand how important the products of chemical industry are in everyday life, try to list five items or materials around you that have NOT been manufactured, processed, or altered from their natural forms.

Based on your list, discuss the following questions.

1 a) Which items on your list were packaged in material that was synthetic?

b) Is the packaging material essential or simply a convenience?

2 In what ways might each item or material on your list be better or worse than a synthetic or processed alternative? Consider factors such as cost, convenience, availability and quality.

45.3 The operation of a chemical process in industry

Fig. 45.4 shows a diagram summarizing a chemical process in industry.

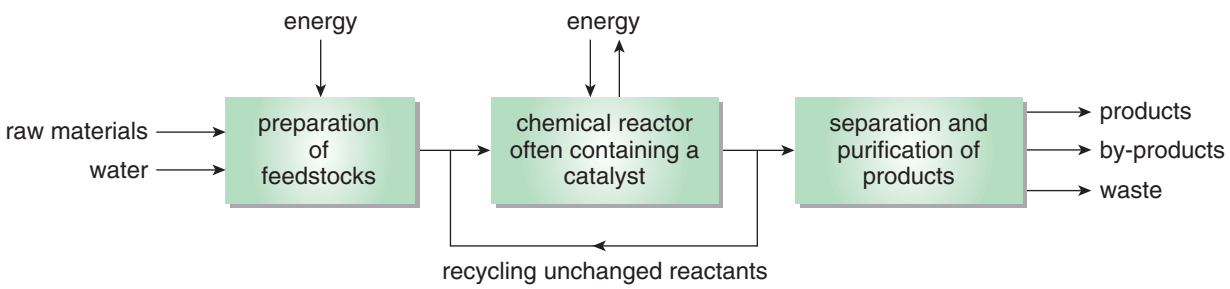


Fig. 45.4 A diagram summarizing a chemical process

The first step in any process is to take the raw materials and convert them into **feedstocks**, that can be fed into the chemical reactor.

Feedstocks are reactants which go into a chemical production process.

Petroleum, for example, is a complex mixture of chemicals. An oil refinery processes chemicals obtained from fractional distillation of petroleum to produce purified feedstocks for chemical synthesis. Table 45.1 shows the feedstocks for the production of some inorganic chemicals.

feedstock 供料

Table 45.1 Feedstocks for the production of some inorganic chemicals

Inorganic chemical	Feedstock(s)
Ammonia	nitrogen, hydrogen
Nitric acid	ammonia, air
Sodium hydroxide	sodium chloride
Chlorine	sodium chloride

The reactants are converted into products in the chemical reactor. The feedstocks may have to be heated or compressed before being fed into the reactor. The reactor often contains a catalyst.

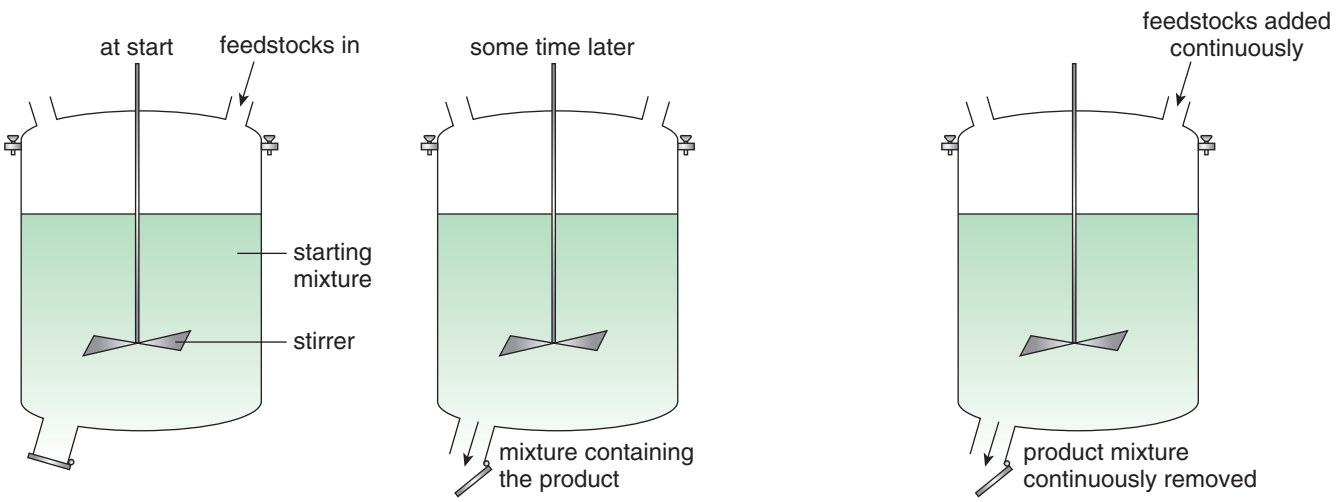
Generally, a mixture of chemicals leaves the reactor. The mixture includes the desired product, but it may include by-products and unchanged reactants as well. So the chemical plant has to include equipment to separate the main product and by-products and to recycle unchanged reactants.

Chemical plants need energy. Some of the chemical reactions occur at a high temperature, so fuels are often needed for the heating. Also, a lot of electrical power is needed for pumps to move reactants and products from one part of the plant to another.

Extension

Batch or continuous?

For a particular process, the sequence of operations may be organized in one of the two ways: *batch* or *continuous*.



(a) Batch reactor (b) Continuous reactor

Fig. 45.5 Comparison of (a) batch and (b) continuous reactors

batch 分批 continuous 連續

Extension

In a batch process, feedstocks are put into a reactor and allowed to react. When the reaction is completed, the product is separated from the reaction mixture. The process is then repeated, batch by batch, until the required amount of product is manufactured (Fig. 45.5a).

In a continuous process, feedstocks are fed in one end of the reactor and product is withdrawn at the other end of the reactor in a continuous flow (Fig. 45.5b).

45.4 Petrochemical industry

Chemicals derived from petroleum or natural gas — petrochemicals* — are an essential part of the chemical industry. Petrochemical industry is a fairly young industry. It only started to grow in the 1940s. During World War II, the demand for synthetic products to replace costly and sometimes less efficient ones caused the industry to develop.

* We have discussed petrochemicals in Topic 7 Fossil Fuels and Carbon Compounds.

Fig. 45.6 shows a kitchen in which some common products (coloured) are made from petrochemicals.

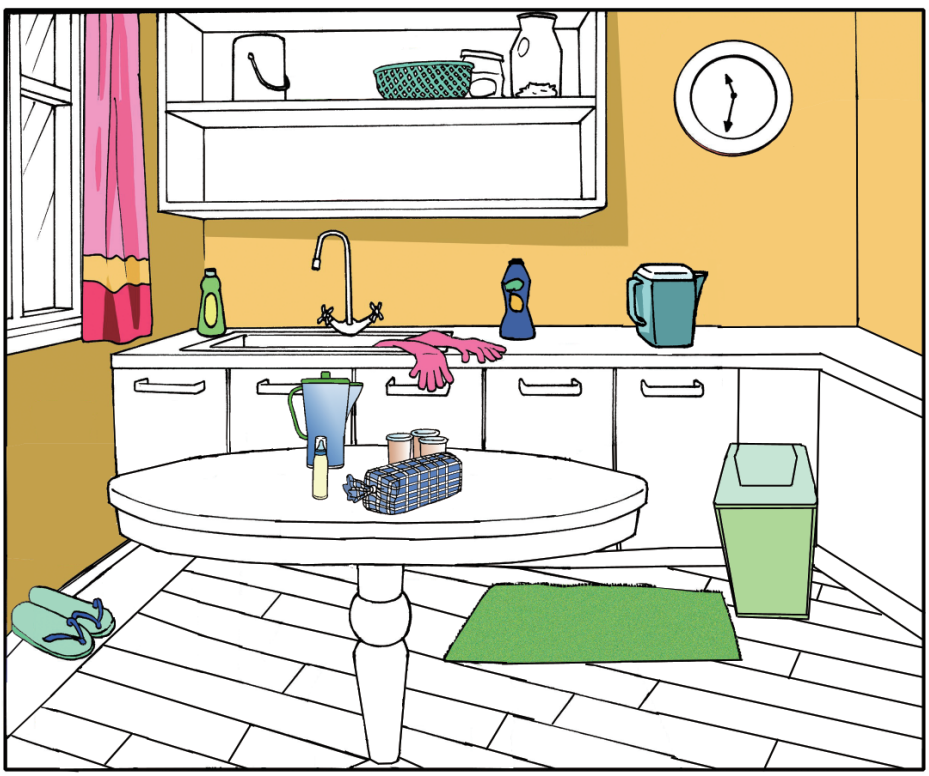


Fig. 45.6 A kitchen in which some common products (coloured) are made from petrochemicals



Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

What are the advantages and disadvantages of using industrial processes for manufacturing products? Discuss with reference to the

- economic;
- environmental; and
- social perspectives.

Discussion
The chemical industry can make a positive contribution to improved quality of life and creation of opportunities for quality employment. However, the chemical industry is often viewed by the general public as causing more harm than good. General ignorance of the end use and value of the industry's products may be a possible reason. For example, petrochemicals do not reach the final consumer. They are first sold to customer industries, undergo several transformations, and then go into products that seem to bear no relation to the initial raw material.
The industry is perceived as being polluting and causing significant environmental damage. Another argument may be based on the concept of 'environmental justice' which implies a fair sharing of the benefits as well as the costs of production. The developing world and low-income areas of the developed world continue to attract a disproportionate share of toxic waste disposal sites and new investment in toxic production facilities.
Finally, the teacher may bring out that the chemical industry in the 21st century needs to address social, economic and environmental concerns simultaneously.

China's petrochemical industry

China was the largest market for chemicals in the world in 2012. The chemicals sales in China in 2012 accounted for one-third of the total sales in the world. One important sector in the chemical industry in China is the production of petrochemicals.

The petrochemical industry contributes to China's economic growth. For example,

- it attracts overseas investments to China;
- setting up of industrial plants provides jobs for local people;
- profit tax from the industry increases the income of the government.

Products of the industry also meet the needs of local consumers.

On the other hand, we need to think about the environmental impact as well. For example,

- the industry may produce toxic waste;
- the consumption of fuels may lead to air pollution and the greenhouse effect;
- water as a coolant in industrial plants may lead to water pollution;
- the cost for removal of chemical waste is high.

China to monitor petrochemical industry for pollution

The State Environmental Protection Administration (SEPA) of China has started to carry out environmental risk assessment on 127 petrochemical key plants along rivers and in areas with dense populations or natural reserves.

A recent benzene spill on the *Songhua river* helped drive the Administration into action. The spill contaminated public water supplies to four million residents of Harbin.

"Recently, a series of serious environmental emergencies, most notably the Songhua river pollution incident, has resulted in great losses to public health, social stability and economic growth" the SEPA Vice Director said.

Already 21 of the plants have been cited for violating environmental protection regulations and faced punishments and further monitoring for compliance. This is to avoid similar environmental incidents in the future.

(Source: JKP 02/2008)

* We will discuss sustainable development further in Unit 48.

Everyone is increasingly aware of the need to protect our world for future generations. This leads to the drive for sustainable development*.

> Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

Chemists face major challenges for developing sustainable chemical processes that respect the environment, improve our quality of life and at the same time are competitive in the market.

The progress in the production of *vitamin C* shows us how the industry improves chemical synthesis so as to solve problems of inadequate supply of natural products and *shrinking* supply of raw materials, as well as minimize the use of toxic solvents and reagents.

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Songhua river 松花江　　vitamin C 維生素 C　　shrinking 縮減的

45.5 Production of vitamin C

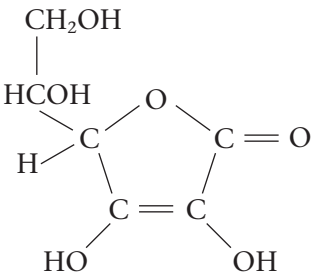
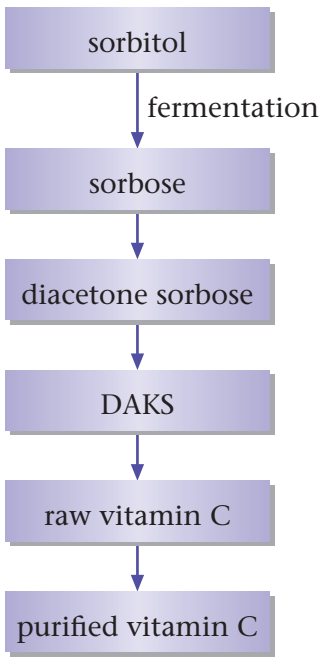


Fig. 45.7 Structure of vitamin C

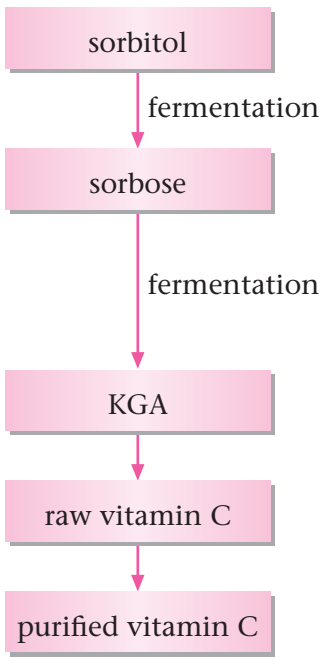
Vitamin C (ascorbic acid) is found in many fruits and vegetables. Although a small amount of vitamin C is extracted from fruits and rose hips, this is not a commercially viable route to meet the demand. Vitamin C is produced from glucose in two main processes: the Reichstein process and the two-step fermentation process.

Extension Reichstein process



DAKS = di-acetone-ketogulonic acid

Two-step fermentation process



KGA = keto-gulonic acid

Fig. 45.8 Processes of producing vitamin C

The Reichstein process was developed in the 1930s. This uses a single fermentation process followed by a purely chemical route.



Fig. 45.9 Vitamin C produced by China

The newer two-step fermentation process was developed in China and is used by all Chinese producers. In this process, a second fermentation step replaces part of the later chemical stages in the Reichstein process.

Compared with the Reichstein process, the two-step fermentation uses less toxic solvents and reagents. There is a reduction in material cost and cost of processing waste products. By introducing improved technology, China has now emerged as the dominant producer of vitamin C in the world.

45.6 The use of catalysts

Catalysts* are involved in the manufacture of most chemicals that we rely on.

* We have discussed catalysts in Topic 10 Rate of Reaction and Topic 12 Patterns in the Chemical World.

A catalyst is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes.

For example, an iron catalyst is used in the synthesis of ammonia in the Haber Process:

$$N_2(g) + 3H_2(g) \xrightleftharpoons{\text{iron catalyst}} 2NH_3(g)$$

Catalysts can reduce the time for production and save energy of a chemical process. Therefore the use of catalysts is important.

How does a catalyst work? To understand that, we need to study the rate of a reaction in greater detail.

Why bother with chemical kinetics?

Some reactions are fast and some are slow. An explosion is a very fast reaction (Fig. 45.10a). On the other hand, limestone weathering is so slow that it takes years before we notice the effect (Fig. 45.10b). *Chemical kinetics* is the study of rates of reaction.

catalyst 催化劑　　chemical kinetics 化學動力學



Fig. 45.10a An explosion is a very fast reaction



Fig. 45.10b Limestone weathering is so slow that it takes years before we notice the effect

We have studied certain aspects of reaction rates in Topic 10 Rate of Reaction. For example, we have learnt about methods for following the progress of a reaction, and used the collision theory to explain the effect of changing concentration and temperature on the rate of a reaction.



Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Suppose you were a chemist working in the research department of a chemical manufacturing company. You are working on the feasibility of producing a new chemical on industrial scale.

Make a list of questions that you would investigate before you can make a decision. Suggest how chemical kinetics would help.

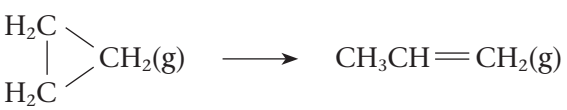
(Hints: You may need to consider the rate of the chemical process involved.)

Discussion
Possible questions:
- What reagents to use?
- How much of each reagent?
- What temperature?
- What pressure?
- Whether or not to use a catalyst?
- How long to allow the reaction to proceed?

Concentration of reagents, temperature, pressure and catalyst are likely to affect the rate of a chemical process. Therefore chemical kinetics is an active research area in the chemical industry.

45.7 The effect of change in concentration on the rate of a reaction

Consider the conversion of cyclopropane to propene:



cyclopropane propene

Table 45.2 shows the change in concentration of cyclopropane at 500 °C.

Table 45.2 The change in concentration of cyclopropane at 500 °C

Time (min)	Concentration of cyclopropane (mol dm^{-3})
0	1.50 (i.e. point A on Fig. 45.11)
5	1.23
10	1.00 (i.e. point B on Fig. 45.11)
15	0.820
20	0.670
25	0.550 (i.e. point C on Fig. 45.11)
30	0.450
35	0.370
40	0.330

Fig. 45.11 shows a plot of concentration of cyclopropane against time.

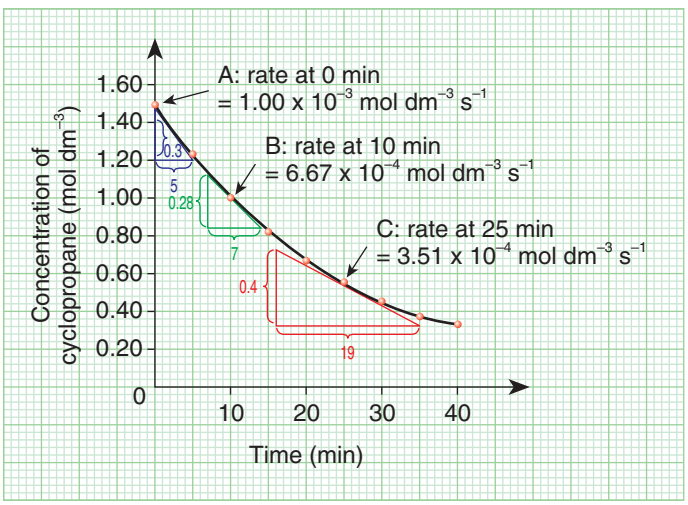


Fig. 45.11 A plot of concentration of cyclopropane against time

From Topic 10 Rate of Reaction, we know that

$$\text{Rate of reaction} = \frac{\text{change in concentration (or amount) of a reactant or a product}}{\text{time}}$$

* We have seen in Topic 10 Rate of Reaction that as it is usual to work with positive reaction rates, we always introduce a minus sign (–) when calculating the rate of disappearance of a reactant.

Thus, for the above reaction, we can express the rate of reaction in terms of the disappearance of cyclopropane: N2

$$\text{Rate} = -\frac{\text{d[cyclopropane(g)]}}{\text{d}t} \ *$$

We can find the rate of reaction at a particular time t from the slope of the tangent to the concentration-time curve at time t.

For example, we can calculate the slope of the tangent at the 10th minute as follows:

$$\text{Slope of tangent at the 10th min} = \frac{(0.84 - 1.12)\ \text{mol dm}^{-3}}{(14 - 7) \times 60\ \text{s}}$$
$$= -6.67 \times 10^{-4}\ \text{mol dm}^{-3}\ \text{s}^{-1}$$

i.e. rate of reaction at the 10th min = 6.67×10^{-4} mol dm^{-3} s^{-1}

Table 45.3 shows reaction rates corresponding to different cyclopropane concentrations at different times.

Table 45.3 Reaction rates corresponding to different cyclopropane concentrations at different times

Time (min)	Concentration of cyclopropane (mol dm^{-3})	Rate of reaction (mol dm^{-3} s^{-1})
0	1.50	1.00×10^{-3}
10	1.00	6.67×10^{-4}
25	0.550	3.51×10^{-4}

Plotting the rate of reaction against the concentration of cyclopropane shown in Table 45.3 gives a straight line (Fig. 45.12).

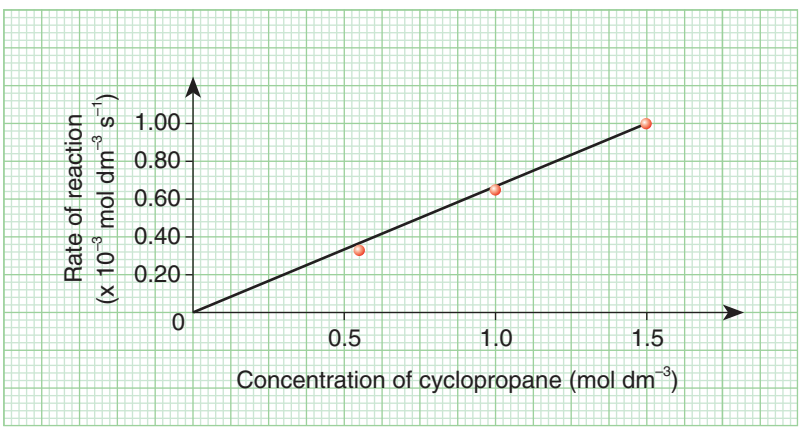


Fig. 45.12 A plot of rate of reaction against the concentration of cyclopropane

N2 The rate of a chemical reaction may be expressed in terms of the rate of disappearance of a reactant or the rate of formation of a product.
For the reaction
A + B ⟶ 2C
2 moles of product C are formed when 1 mole of A reacts with 1 mole of B.
∴ rate of formation of C
= 2 x rate of disappearance of A or B

The graph in Fig. 45.12 shows that the rate of reaction is directly proportional to the concentration of cyclopropane.

Rate $\propto$ concentration of cyclopropane

i.e. Rate $\propto$ [cyclopropane(g)]
Rate = k[cyclopropane(g)]

where k is a constant, known as the **rate constant** for the reaction.

An expression of this kind is called **rate equation**. It relates the rate of a reaction to the rate constant and the concentration(s) of the reactant(s).

45.8 Order of reaction

For a general reaction of the type:

$aA + bB \longrightarrow$ products

the rate equation takes the form

Rate = $k[A]^x[B]^y$

The index 'x' is the order of reaction with respect to the reactant A while the index 'y' is the order of reaction with respect to the reactant B. The overall order of reaction is $x + y$.

Notice that x and y are constants for that particular reaction. We can only determine them experimentally. We cannot deduce them from the coefficients of the chemical equation of the reaction. N3

For example, the gas dinitrogen oxide (N_2O) decomposes on a heated gold surface according to the following equation:

$$2N_2O(g) \xrightarrow{Au} 2N_2(g) + O_2(g)$$

Experiments show that the rate of reaction is directly proportional to the concentration of dinitrogen oxide (Table 45.4).

N3 Some experimentally determined rate equations
Some experimentally determined rate equations are shown below.

Reaction	Chemical equation	Rate equation
1	$H_2(g) + I_2(g) \longrightarrow 2HI(g)$	Rate = $k[H_2(g)][I_2(g)]$
2	$NO(g) + CO(g) + O_2(g) \longrightarrow NO_2(g) + CO_2(g)$	Rate = $k[NO(g)]^2$
3	$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l)$	Rate = $k[BrO_3^-(aq)][Br^-(aq)][H^+(aq)]^2$

We can see from the rate equation of Reaction 1 that the rate of reaction is proportional to the concentrations of both $H_2(g)$ and $I_2(g)$. However, for Reaction 2, $CO(g)$ and $O_2(g)$ do not appear in the rate equation, even though they are present in the reactant side of the chemical equation. From Reaction 3, we can see that there is no relationship between the values of the order indices and the coefficients of the chemical equation.

rate constant 速率常數　　rate equation 速率方程式

Table 45.4 Relative rates of decomposition of dinitrogen oxide corresponding to different concentrations of dinitrogen oxide

Concentration of N_2O (mol dm^{-3})	Relative rate of decomposition of N_2O
1.00	8
0.500	4
0.250	2
0.125	1

When the concentration doubles, so does the decomposition rate.

Rate $\propto$ $[N_2O(g)]$

Thus, the rate equation is

Rate = $k[N_2O(g)]$

which can be written as

Rate = $k[N_2O(g)]^1$

The reaction is first order with respect to dinitrogen oxide.

Zero, first and second order reactions

Zero order reactions

For a zero order reaction:

A $\longrightarrow$ product(s)

the rate equation is

Rate = $k[A]^0 = k$

(since anything to the power 0 equals 1).

Thus, the rate of a zero order reaction is a constant, independent of the concentration of reactant.

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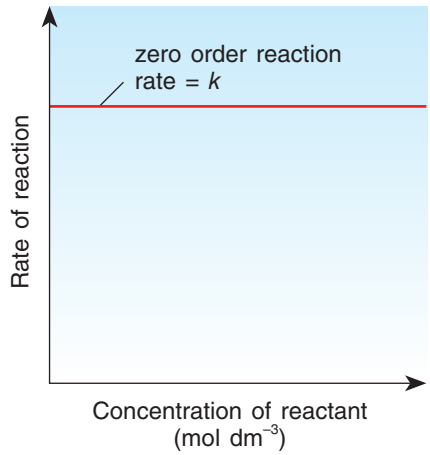


Fig. 45.13 A rate-concentration graph for a zero order reaction

When we plot the rate of reaction against the concentration of reactant for a zero order reaction, a horizontal line is obtained (Fig. 45.13).

$$\text{Rate} = k[\text{reactant}]^0 = k$$

For example, ammonia gas decomposes on a hot tungsten wire according to the following equation:

$$2NH_3(g) \xrightarrow{W} N_2(g) + 3H_2(g)$$

The rate of decomposition does not depend upon the concentration of ammonia. The rate is fixed. Thus,

$$\text{Rate} = k[NH_3(g)]^0$$

First order reactions

In a first order reaction of the type:

$$A \longrightarrow \text{product(s)}$$

the rate equation is

$$\text{Rate} = k[A]$$

The rate of the above reaction is directly proportional to the concentration of reactant A.

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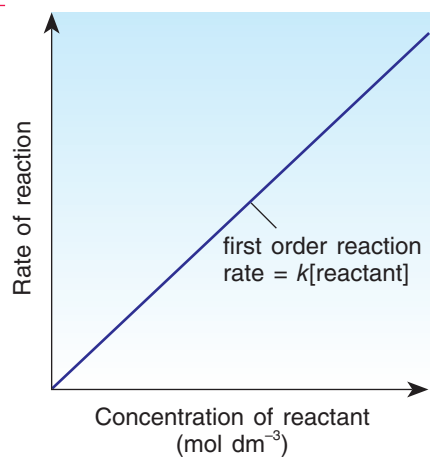


Fig. 45.14 A rate-concentration graph for a first order reaction

When we plot the rate of reaction against the concentration of reactant for a first order reaction, a straight line with a constant gradient is obtained (Fig. 45.14). Since rate = k[reactant], the gradient of the straight line is equal to the rate constant k.

For example, hydrogen peroxide (H_2O_2) decomposes in the presence of a catalyst according to the following equation:

$$2H_2O_2(aq) \xrightarrow{\text{catalyst}} 2H_2O(l) + O_2(g)$$

The decomposition of hydrogen peroxide is a first order reaction. The rate of decomposition is directly proportional to the concentration of the hydrogen peroxide solution.

$$\text{Rate} = k[H_2O_2(aq)]$$

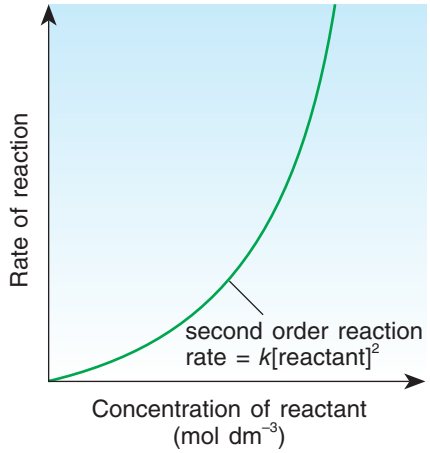


Fig. 45.15 A rate-concentration graph for a second order reaction

Second order reactions

The simplest type of second order reaction involves only one reactant:

$$A \longrightarrow \text{product(s)}$$

the rate equation is

$$\text{Rate} = k[A]^2$$

The rate of the above reaction is directly proportional to the square of the concentration of A.

When we plot the rate of reaction against the concentration of reactant for such a second order reaction, a curved line is obtained (Fig. 45.15). Compare Fig. 45.13 – Fig. 45.15. Notice that for reactions of different orders, the concentration of reactant affects the rate of reaction differently.

For example, ethanal vapour (CH_3CHO) decomposes at 550 °C according to the following equation:

$$CH_3CHO(g) \longrightarrow CH_4(g) + CO(g)$$

The decomposition of ethanal is a second order reaction. The rate of decomposition is directly proportional to the square of the concentration of ethanal.

$$\text{Rate} = k[CH_3CHO(g)]^2$$

Another type of second order reaction involves two reactants:

$$A + B \longrightarrow \text{product(s)}$$

and the rate equation is

$$\text{Rate} = k[A][B]$$

The reaction is first order with respect to A and first order with respect to B. So, the overall order of reaction is 2.

For example, hydrogen gas and iodine vapour react to give hydrogen iodide according to the following equation:

$$H_2(g) + I_2(g) \longrightarrow 2HI(g)$$

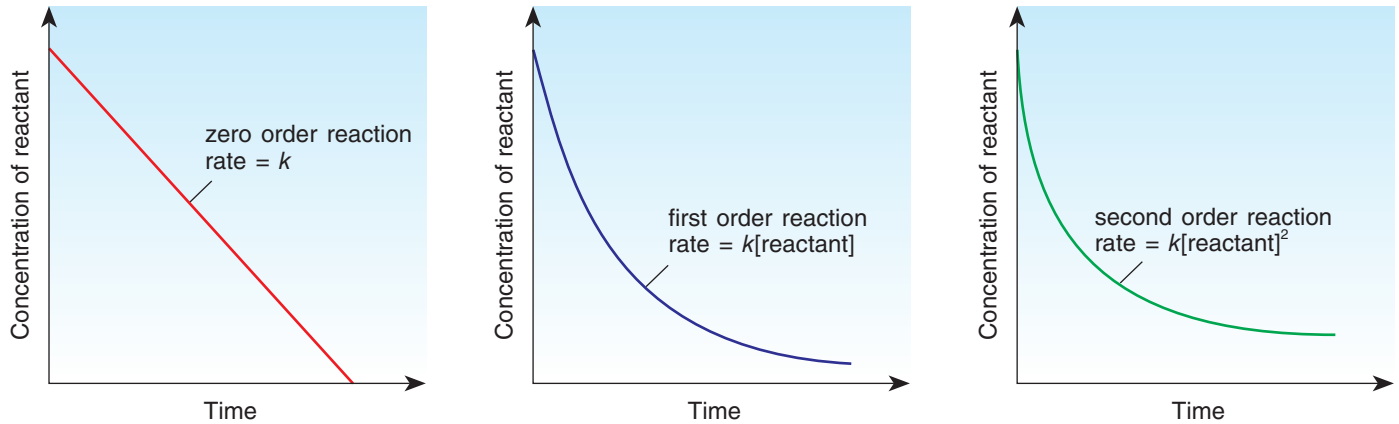
The rate of reaction is directly proportional to the concentration of hydrogen and iodine separately.

$$\text{Rate} = k[H_2(g)][I_2(g)]$$

Thus, the overall order of reaction is 2.

Concentration-time graph

Fig. 45.16 shows how the reactant concentration varies with time for zero, first and second order reactions. Notice that in all the cases the reactant concentration decreases as the reaction proceeds.



(a) zero order reaction (b) first order reaction (c) second order reaction

Fig. 45.16 Concentration-time graph

When a reaction is zero order with respect to a reactant, the reactant concentration has no effect on the rate of the reaction. The concentration-time graph for a zero order reaction shows a constant decline in concentration as the reaction proceeds (Fig. 45.16a).

45.9 Units of rate constant *k* for reactions of different order

We have seen that

$$\text{Rate} = \frac{\text{change in concentration (or amount) of a reactant or a product}}{\text{time}}$$

and that rate is measured in mol dm^{-3} s^{-1}.

For a zero order reaction, where rate = $k[\text{reactant}]^0$ and $[\text{reactant}]^0 = 1$, rate constant k = rate.

So, the units of k for a zero order reaction are those of rate, i.e. mol dm^{-3} s^{-1}.

For a first order reaction where rate = k[reactant], we can write

$$k = \frac{\text{rate}}{[\text{reactant}]} \text{ or } k = \frac{\text{rate}}{\text{concentration}}$$

So, k is in $\frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{\text{mol dm}^{-3}}$, i.e. s^{-1}

Thus, for a first order reaction, the units of k are s^{-1}.

For a second order reaction, where rate = k[reactant]^2, we can write

$$k = \frac{\text{rate}}{[\text{reactant}]^2}$$

Alternatively, rate = k[reactant A][reactant B], giving

$$k = \frac{\text{rate}}{[\text{reactant A}][\text{reactant B}]}$$

In both cases, we have

$$k = \frac{\text{rate}}{(\text{concentration})^2}$$

So, k is in $\frac{\text{mol dm}^{-3}\ \text{s}^{-1}}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$, i.e. $\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$*

Thus, for a second order reaction, the units of k are $\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$. Similar rules apply for higher order reactions.

* Notice that dm^3 is placed first when giving the units. This follows the convention of placing a unit raised to a positive power first.

Table 45.5 Units of rate constant k for reactions of different order

Overall order of reaction	Expression for rate constant k	Units of rate constant k
0	k = rate	$\text{mol dm}^{-3}\ \text{s}^{-1}$
1	$k = \frac{\text{rate}}{\text{concentration}}$	s^{-1}
2	$k = \frac{\text{rate}}{\text{concentration}^2}$	$\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$

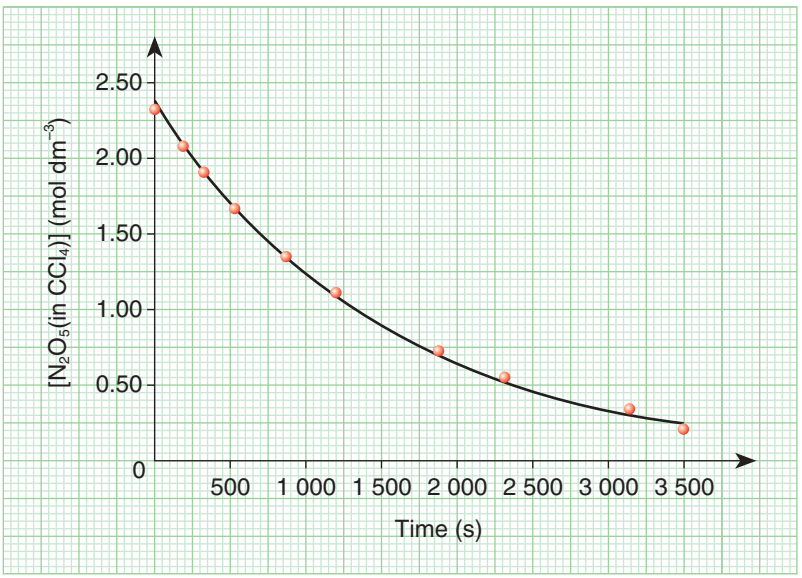
Example

45.1 Determining the rate equation for the decomposition of dinitrogen pentoxide

The decomposition of dinitrogen pentoxide in tetrachloromethane can be represented by the following equation:

$$2N_2O_5(\text{in } CCl_4) \longrightarrow 4NO_2(\text{in } CCl_4) + O_2(g)$$

An experiment was carried out to study the rate of the decomposition of dinitrogen pentoxide at 318 K. The results obtained are shown in the graph below.



a) Find the rate of reaction at FOUR different times by drawing tangents at four points on the graph, and then calculating the slope at each point.

b) Plot a graph of rate of reaction against $[N_2O_5(\text{in } CCl_4)]$. What order of reaction with respect to $[N_2O_5(\text{in } CCl_4)]$ is suggested by this graph?

c) Calculate the rate constant by finding the slope of the graph plotted in (b).

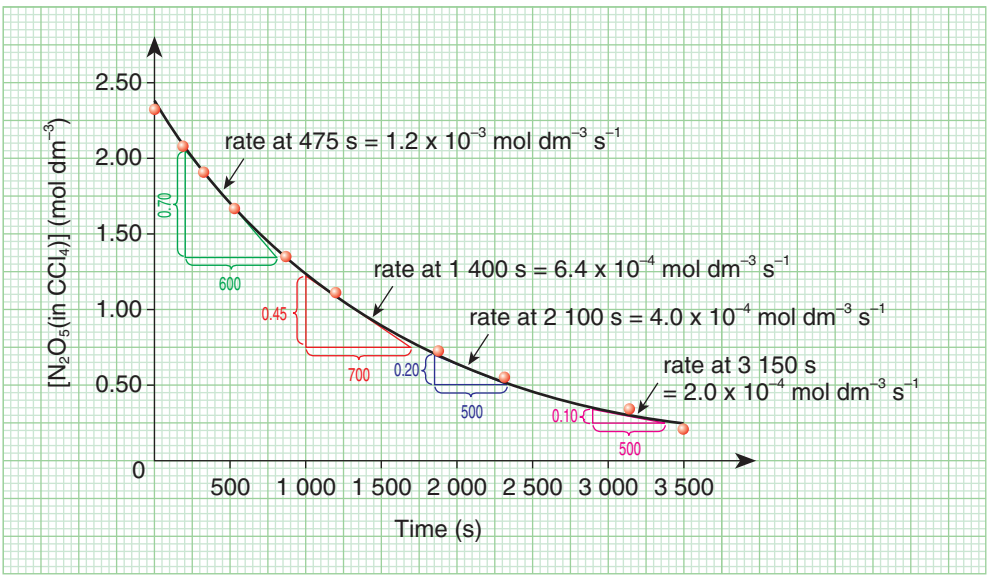
d) Write the rate equation for the reaction.

AL 2003 Paper 2 Q3(a)(ii),(iii)
AL 2004 Paper 2 Q3(a)(ii)
AL 2006 Paper 2 Q3(a)(ii)

Solution

a) We can express the rate of reaction in terms of the disappearance of dinitrogen pentoxide.

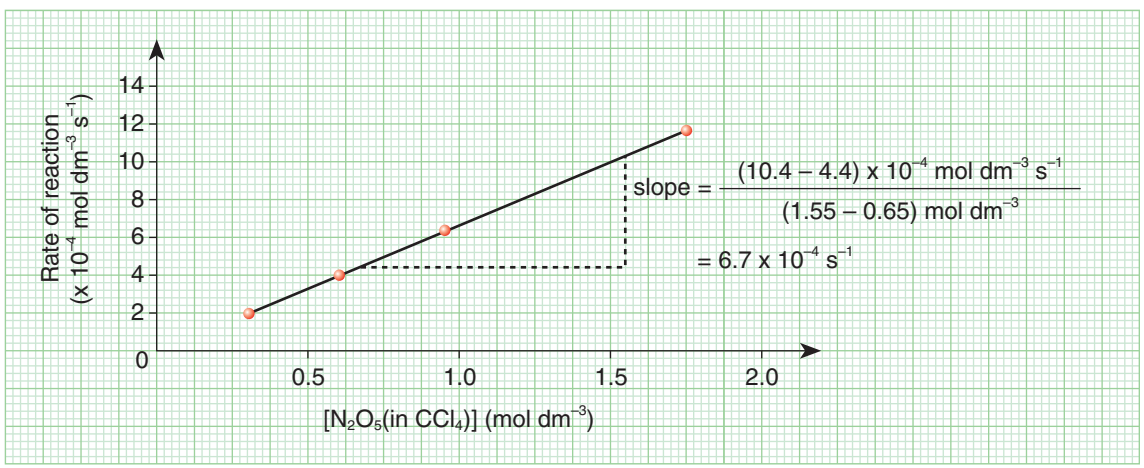
$$\text{Rate} = -\frac{d[N_2O_5(\text{in } CCl_4)]}{dt}$$



The reaction rates corresponding to different $[N_2O_5(\text{in } CCl_4)]$ are as follows:

Time(s)	$[N_2O_5(\text{in } CCl_4)]$ (mol dm^{-3})	Rate of reaction (mol dm^{-3} s^{-1})
475	1.75	1.2×10^{-3}
1 400	0.95	6.4×10^{-4}
2 100	0.60	4.0×10^{-4}
3 150	0.30	2.0×10^{-4}

b)



The graph shows that the rate of reaction is directly proportional to the concentration of N_2O_5(in CCl_4).

i.e. Rate $\propto$ $[N_2O_5(\text{in } CCl_4)]$

$\therefore$ the reaction is first order with respect to dinitrogen pentoxide.

c) Slope of the graph plotted in (b) = $k = \dfrac{(10.4 - 4.4) \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}}{(1.55 - 0.65) \text{ mol dm}^{-3}}$

$= 6.7 \times 10^{-4} \text{ s}^{-1}$

d) Rate = $k[N_2O_5(\text{in } CCl_4)]$

where k is the rate constant.

Checkpoint

1 Consider the rate equations for the following reactions.

a) $4HBr(g) + O_2(g) \longrightarrow 2H_2O(g) + 2Br_2(g)$

Rate = $k[HBr(g)][O_2(g)]$

b) $2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$

Rate = $k[NO(g)]^2[O_2(g)]$

c) $NO_2(g) + CO(g) \longrightarrow NO(g) + CO_2(g)$

Rate = $k[NO_2(g)]^2$

For each reaction,

i) give the order of reaction with respect to each reactant;

ii) state the overall order of reaction.

DSE 2015 Paper 2 Q1(a)(i)(2)

1 a) i) The reaction is first order with respect to HBr(g).
The reaction is first order with respect to O_2(g).
ii) The overall order of reaction is 2.
b) i) The reaction is second order with respect to NO(g).
The reaction is first order with respect to O_2(g).
ii) The overall order of reaction is 3.
c) i) The reaction is second order with respect to NO_2(g).
ii) The overall order of reaction is 2.

2 Write down the rate equation for each of the following reactions.

a) The reaction of 1-chlorobutane with sodium hydroxide solution is first order with respect to 1-chlorobutane and first order with respect to hydroxide ion.

$C_4H_9Cl(l) + OH^-(aq) \longrightarrow C_4H_9OH(aq) + Cl^-(aq)$

Rate = $k[C_4H_9Cl(l)][OH^-(aq)]$

b) The hydrolysis of sucrose ($C_{12}H_{22}O_{11}$) is first order with respect to sucrose and first order with respect to acid catalyst ($H^+(aq)$).

$C_{12}H_{22}O_{11}(aq) + H_2O(l) \longrightarrow 2C_6H_{12}O_6(aq)$

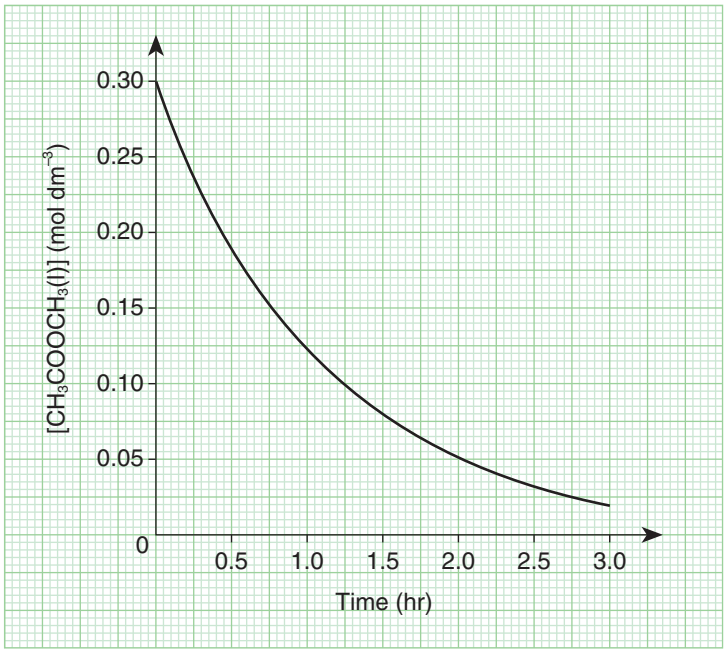
Rate = $k[C_{12}H_{22}O_{11}(aq)][H^+(aq)]$

3 The ester methyl ethanoate reacts with water according to the following equation:
Refer to page T15.

$$CH_3COOCH_3(l) + H_2O(l) \longrightarrow CH_3COOH(aq) + CH_3OH(aq)$$

An experiment to investigate the rate of this hydrolysis showed that the concentration of methyl ethanoate varied as shown in the graph below.

The water in this reaction was not only a reactant but was also the solvent. It was in large excess.



a) Suggest a method by which the progress of the reaction can be followed.

b) Find the rate of reaction at FOUR different times.

c) Plot a graph of rate of reaction against $[CH_3COOCH_3(l)]$. What order of reaction with respect to $[CH_3COOCH_3(l)]$ is suggested by this graph? 1

d) Calculate the rate constant from the graph plotted in (c). $0.846\ hr^{-1}$

e) Write the rate equation for the reaction.

45.10 Experimental determination of the rate equation for a reaction — the method of initial rate N4

N4 The progress curve method for determining the rate equation for a reaction
A progress curve shows how the concentration of a reactant (or a product) changes as the reaction proceeds. Refer to the conversion of cyclopropane to propene under Section 45.7 for the method of finding the rate of the reaction at different concentrations using the progress curve. Then the order of the reaction with respect to cyclopropane can be found.

To determine the rate equation for a reaction, we must do experiments. Most reactions involve more than one reactant, and in this case, we have to do several experiments to find the order with respect to each reactant separately.

To find the order of reaction with respect to a certain reactant, control the variables so that only the concentration of that reactant varies. All the measurements should be taken at the same temperature.

A way of controlling the concentration of another reactant is to have a large excess of it, so that over the course of the experiment, its concentration does not change significantly (refer to Example 45.2). N5

N5 Refer to page T17.

The initial rate of reaction is the rate of reaction at the start.

DSE 2014 Paper 2 Q1(b)(i)

One way to find the order of reaction with respect to a certain reactant is to investigate how the initial rate depends on the initial concentration of the reactant — the method of initial rate. N6

N6 Examination questions often give the initial reactant concentrations and initial rate of a reaction and ask students to derive the rate equation for the reaction.

In general, it is preferable to measure the initial rate because as the reaction proceeds, concentrations of the reactants decrease and it may become difficult to measure the changes accurately.

DSE 2012 Paper 2 Q1(c)(i)

To find the order of reaction with respect to a certain reactant, we need to run the same experiment several times, but using a different concentration of the reactant. Concentrations of all the other reactants and reaction conditions are kept the same each time. Data of initial concentrations and initial rates allow us to find the order with respect to the reactant. For a reaction involving more than one reactant, we need to apply the same procedure to each reactant in turn.

Determining initial rates from different concentration-time curves

Look at the decomposition of hydrogen peroxide solution. This reaction proceeds slowly under normal conditions, but it is greatly speeded up by catalysts. A particularly effective catalyst is the enzyme catalase.

$$2H_2O_2(aq) \xrightarrow{\text{catalase}} 2H_2O(l) + O_2(g)$$

We can use the experimental set-up shown in Fig. 45.17 to investigate the enzyme-catalyzed decomposition of hydrogen peroxide solution.

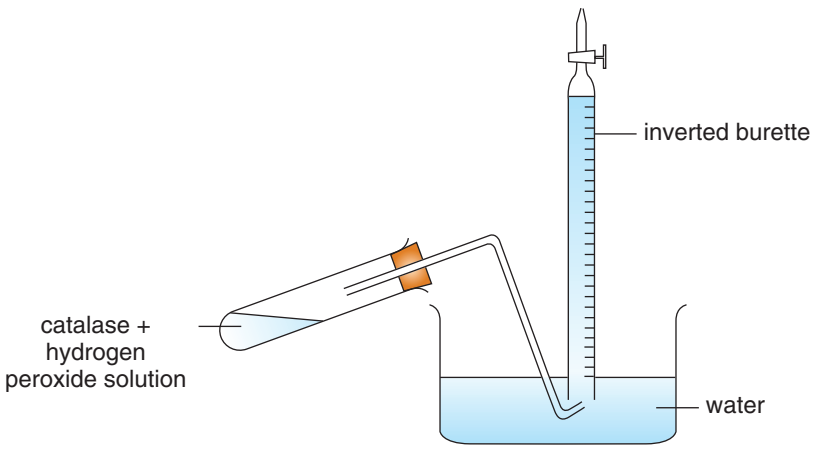


Fig. 45.17 Experimental set-up for investigating the enzyme-catalyzed decomposition of hydrogen peroxide solution

We can measure the total volume of oxygen given off at different times from the start of the experiment and plot a graph of this against time. N7

Fig. 45.18 shows a graph of results obtained by starting with a hydrogen peroxide solution of 0.40 mol dm^{-3}.

N7 Following the progress of the decomposition of hydrogen peroxide solution via titration

The concentration of $H_2O_2(aq)$ in the reaction mixture at different times can be determined by titrating against standard $KMnO_4(aq)$.

$2MnO_4^-(aq) + 5H_2O_2(aq) + 6H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$

The steps are as follows:

- Withdraw a known volume of the reaction mixture at regular time intervals.
- Add the reaction mixture to a large excess of dilute acid.
- Titrate against standard $KMnO_4(aq)$ until a pale pink colour appears (redox titration). The volume of $KMnO_4(aq)$ used is proportional to the concentration of $H_2O_2(aq)$ present.

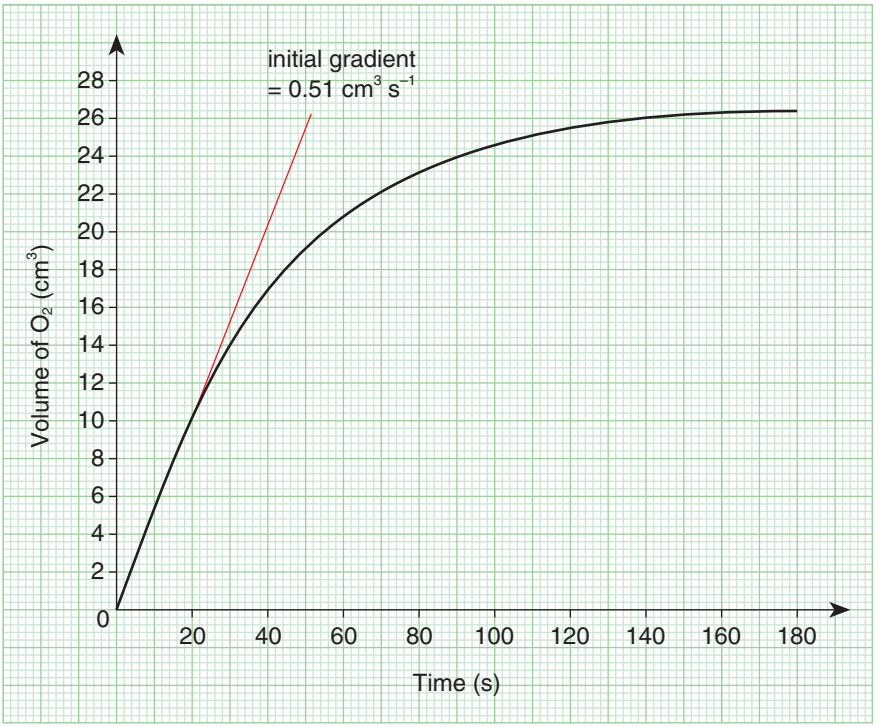


Fig. 45.18 A graph of results obtained by starting with a hydrogen peroxide solution of concentration 0.40 mol dm^{-3}

We can find the initial rate by drawing a tangent to the curve at time zero and measuring the slope of this tangent.

In the example shown in Fig. 45.18,

$$\text{slope of tangent to curve at time zero} = \frac{(25.5 - 0)\ \text{cm}^3}{(50 - 0)\ \text{s}}$$
$$= 0.51\ \text{cm}^3\ \text{s}^{-1}$$

So, the initial rate of formation of oxygen is 0.51 cm^3 s^{-1}.

To find the order of reaction with respect to hydrogen peroxide solution, we can repeat the experiment using hydrogen peroxide solutions of different concentrations. The concentration of the enzyme catalase and all other conditions such as temperature are kept the same each time.

Fig. 45.19 shows results obtained in a certain experiment. As we will expect, the curves start off with differing gradients, depending on the initial concentration of the hydrogen peroxide solution.

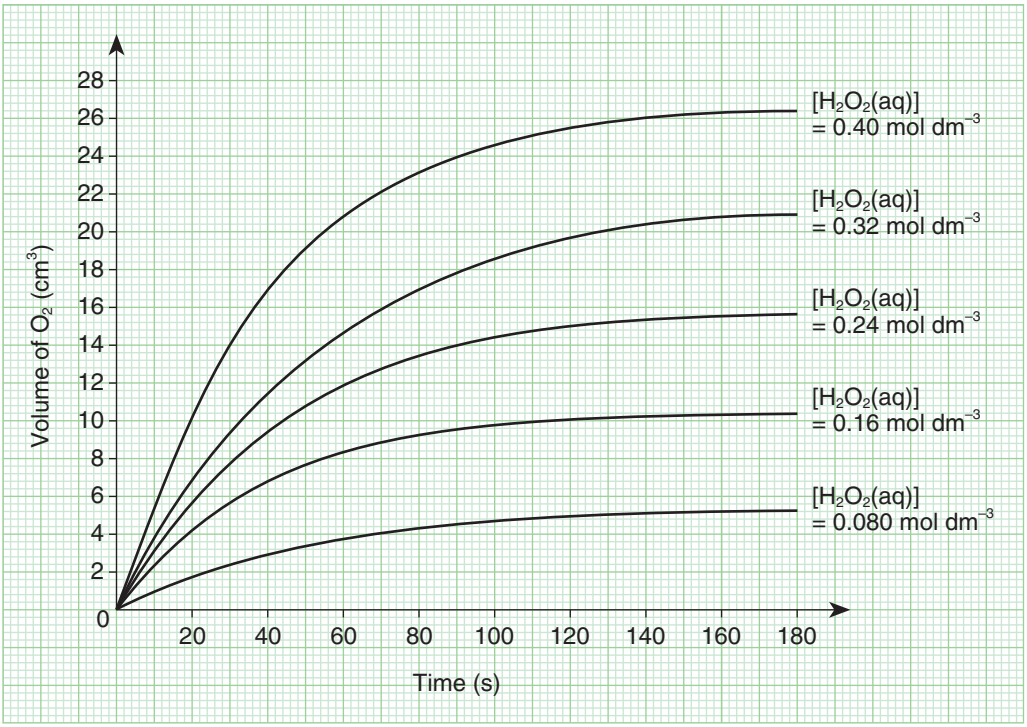


Fig. 45.19 Results obtained for decomposition of hydrogen peroxide solutions of different initial concentrations

Table 45.6 shows the initial rates of formation of oxygen corresponding to hydrogen peroxide solutions of different initial concentrations.

Table 45.6 The initial rates of formation of oxygen corresponding to hydrogen peroxide solutions of different initial concentrations

Initial concentration of hydrogen peroxide solution (mol dm^{-3})	Initial rate of formation of oxygen (cm^3 s^{-1})
0.40	0.51
0.32	0.41
0.24	0.32
0.16	0.21
0.080	0.10

Fig. 45.20 shows a plot of initial rate of formation of oxygen against initial concentration of hydrogen peroxide solution. It is a straight line. This means that the rate is directly proportional to the concentration of hydrogen peroxide solution. In other words,

$$\text{Rate} \propto [H_2O_2(aq)]$$

or $$\text{Rate} = k[H_2O_2(aq)]$$

Thus, the reaction is first order with respect to hydrogen peroxide solution.

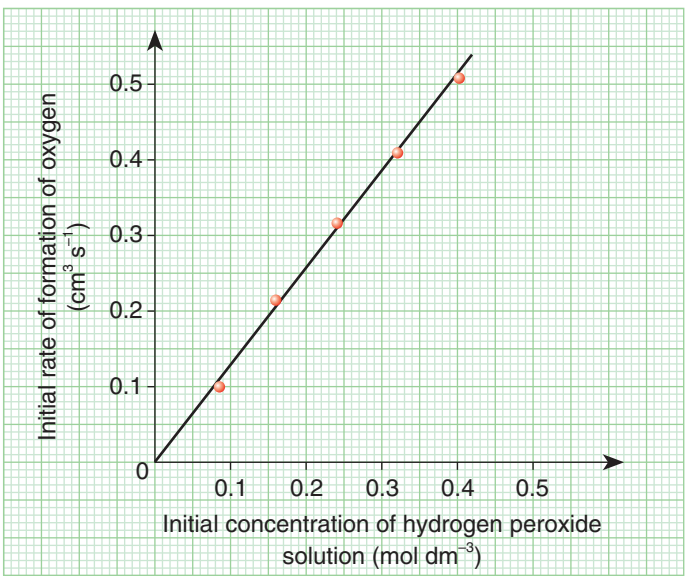


Fig. 45.20 A plot of initial rate of formation of oxygen against initial concentration of hydrogen peroxide solution

If we want to look at the effect of varying the concentration of catalase on the rate, we must keep the concentration of hydrogen peroxide solution constant when we carry out the experiment.

Carry out several experiments with different initial concentrations of catalase, and measure the initial rates of formation of oxygen. Then we can find the order of the reaction with respect to catalase. N8

N8 The rate of the reaction is also proportional to the concentration of catalase. The rate equation is
Rate = $k[H_2O_2(aq)][\text{catalase}]$

Reference Website

- Refer to the following website for a virtual experiment for determining the rate equation and the rate constant of the reaction between $H_2SeO_3(aq)$ and $I^-(aq)$ ion:
http://www.chm.davidson.edu/vce/Kinetics/MethodOfInitialRates.html
(accessed May 2015)
- Refer to the following website for a virtual experiment for determining the rate equation and the rate constant of the reaction between $BrO_3^-(aq)$ ion and $Br^-(aq)$ ion:
http://www.chm.davidson.edu/vce/Kinetics/BromateBromideReaction.html
(accessed May 2015)

Using the reciprocal of the time required to produce a small amount of a product as a measure of the initial rate

Another way of measuring the initial rate of a reaction is to measure how long the reaction takes to produce a small, fixed amount of one of the products.

A good example of this method is the reaction between sodium thiosulphate solution and dilute sulphuric acid:

$$Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$$

* We have discussed this method of following the progress of the reaction in Topic 10 Rate of Reaction.

As the reaction proceeds, sulphur forms and this changes the light transmittance of the reaction mixture. We can place the reaction vessel over a cross on a piece of white paper and view from above as shown in Fig. 45.21. We can no longer see the cross when enough sulphur has formed and the solution becomes *opaque**.



(a) When the reactants are being mixed, the cross can be clearly seen.

(b) The cross gets fainter as more sulphur precipitate forms.

(c) The cross can no longer be seen when enough sulphur has formed.

Fig. 45.21 Different stages of the reaction between sodium thiosulphate solution and dilute sulphuric acid

To find the order of reaction with respect to sodium thiosulphate solution, we can measure the time to reach the opaque stage using different initial concentrations of sodium thiosulphate solution. The total volume of each reaction mixture and the concentration of the sulphuric acid must be kept constant in each experiment.

Notice that the amount of sulphur needed to obscure the cross is the same in each experiment. That means we are measuring the time taken to produce a small, fixed amount of sulphur in each experiment.

If the reaction is fast, the time to reach the opaque stage will be short. If the reaction is slow, the time will be long. The time interval taken to reach such a stage is thus inversely proportional to the average rate of reaction from start to the opaque stage.

For each sodium thiosulphate solution,

$$\text{average rate from start to the opaque stage} \propto \frac{1}{\text{time to reach the opaque stage}}$$

opaque 不透明的

When the opaque stage is reached, only a small amount of sulphur has formed and the reaction has only proceeded a small part of the way to completion. We can take the average rate from start to the opaque stage to be approximately equal to the initial rate.

$$\text{Initial rate} \propto \frac{1}{\text{time to reach the opaque stage}}$$

Table 45.7 shows results obtained in a certain experiment.

Table 45.7 Results of experiments involving the reaction between sodium thiosulphate solution and dilute sulphuric acid

Experiment no.	Volume of reagent used (cm^3)			Time to reach opaque stage (t) (s)	Relative initial rate ($\frac{1}{t}$) (s^{-1})
	0.5 mol dm^{-3} $H_2SO_4(aq)$	$H_2O(l)$	0.10 mol dm^{-3} $Na_2S_2O_3(aq)$		
1	20.0	25.0	5.0	181	0.0055
2	20.0	20.0	10.0	91	0.011
3	20.0	15.0	15.0	61	0.016
4	20.0	10.0	20.0	45	0.022
5	20.0	5.0	25.0	36	0.028
6	20.0	0.0	30.0	30	0.033

Notice that water is added in some cases to keep the total volume of all reaction mixtures constant. The same volume of sulphuric acid is used in each experiment so that the only variable is the volume of sodium thiosulphate solution used, i.e. the initial concentration of the solution.

Fig. 45.22 shows a plot of relative initial rate ($\frac{1}{t}$) against the volume of sodium thiosulphate solution used. A straight line is obtained.

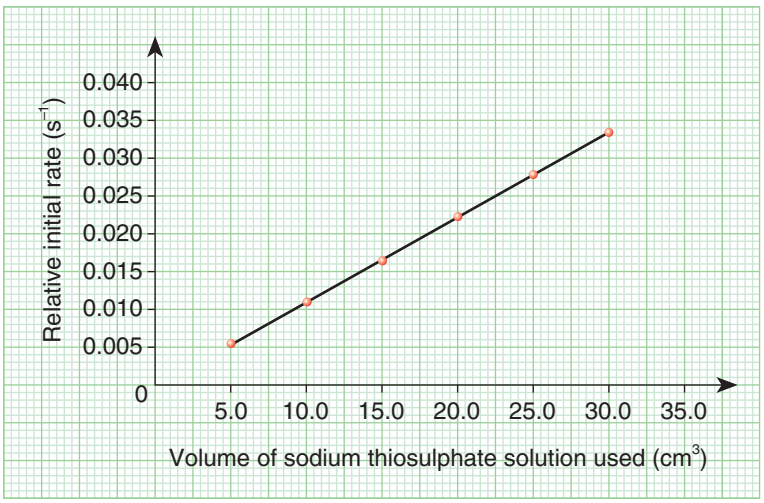


Fig. 45.22 A plot of relative initial rate ($\frac{1}{t}$) against the volume of sodium thiosulphate solution used

As the volume of sodium thiosulphate solution used is proportional to the concentration of the solution in the reaction mixture, we can conclude that the rate is directly proportional to the concentration of sodium thiosulphate solution.

$$\text{Rate} \propto [Na_2S_2O_3(aq)]$$

$$\text{or Rate} = k[Na_2S_2O_3(aq)]$$

Thus, the reaction is first order with respect to sodium thiosulphate solution.

In the following activity, you are going to determine the rate equation for the reaction between sodium thiosulphate solution and dilute sulphuric acid using the method of initial rate.

Video

Investigating the effect of varying the concentration of sodium thiosulphate solution on the rate of its reaction with dilute sulphuric acid

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 45.1

Determining the rate equation for the reaction between sodium thiosulphate solution and dilute sulphuric acid using a microscale chemistry experiment.

45.11 Using the iodine clock method to determine the rate equation for a reaction

Peroxodisulphate ion ($S_2O_8^{2-}$) and iodide ion (I^-) react together in solution to form sulphate ion and iodine according to the following equation:

$$S_2O_8^{2-}(aq) + 2I^-(aq) \longrightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

Let us call this the main reaction.

We can detect the presence of iodine in a reaction mixture by adding some starch to the reaction mixture. Iodine forms a dark blue complex with starch.

$$\text{starch solution} + I_2(aq) \longrightarrow \text{dark blue complex}$$

One way of measuring the initial rate of the reaction between peroxodisulphate ion and iodide ion (i.e. the main reaction) is to measure the reaction time taken for a small, fixed amount of iodine to form. We can measure this time by adding a small, fixed amount of thiosulphate ion and starch solution to the reaction mixture at the start. The thiosulphate ion will consume any iodine formed from the main reaction.

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

As long as there is thiosulphate ion in the reaction mixture, any iodine formed from the main reaction will be consumed by the thiosulphate ion immediately. The starch will not be affected. N9

N9 Refer to page T18.

What we see is a colourless solution mixture sitting there as though nothing is happening; then suddenly, it turns dark blue. This is the instant at which all the thiosulphate ion has been used up. Free iodine in the reaction mixture gives a dark blue complex with the starch.

Reference Website

Refer to the following website for a virtual experiment studying the rate kinetics of the reaction between $S_2O_8^{2-}(aq)$ ion and $I^-(aq)$ ion using the iodine clock:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/kinetics2/iodine_clock.html
(accessed May 2015)

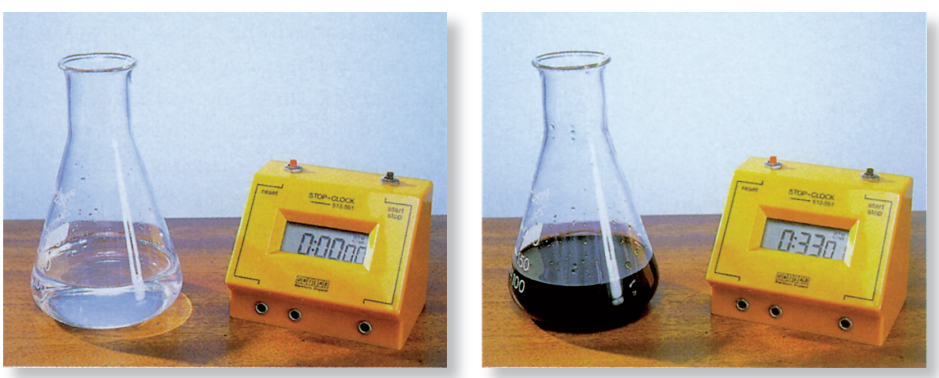
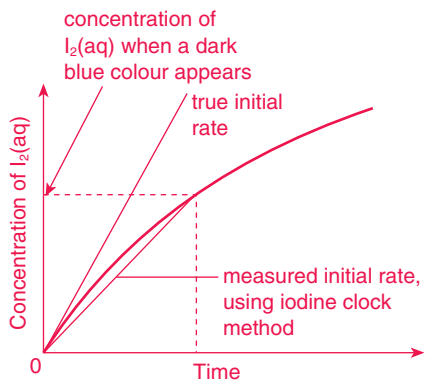


Fig. 45.23 An iodine clock experiment — the reaction mixture turns dark blue when all the thiosulphate ion has been used up

N10 An approximate measure of the initial rate using the iodine clock method

The initial rate measured using the iodine clock method is an approximate value because it assumes that the concentration-time graph is a straight line not only near the origin, but up to the point when the dark blue colour appears. The graph below shows that the iodine clock method gives a value for the initial rate that is lower than the true value.



If we measure how long it takes for the dark blue colour to appear, we can find how long it takes to form enough iodine to use up all the thiosulphate ion, and thus how long it takes to form a certain amount of iodine.

We may use the reciprocal of the time for the dark blue colour to appear as an approximate measure of the initial rate. N10

$$\text{Initial rate} \propto \frac{1}{\text{time for a dark blue colour to appear}}$$

The approximation is reasonable provided that the reaction has only proceeded a small part of the way to completion.

To find the order of reaction with respect to iodide ion, we can measure the time taken for a dark blue colour to appear using different initial concentrations of iodide ion in solution. The total volume of the reaction mixture, and concentrations of the peroxodisulphate ion and thiosulphate ion in solution must be kept constant in each experiment.

This method of study is sometimes called the clock method and this experiment is an example of an **iodine clock experiment**.



45.2 Deducing the order of reaction with respect to a reactant

A student investigated the rate of the reaction between bromine and methanoic acid in aqueous solution. The reaction is acid catalyzed.

$$Br_2(aq) + HCOOH(aq) \xrightarrow{H^+} 2Br^-(aq) + 2H^+(aq) + CO_2(g)$$

The initial concentration of bromine was 0.0100 mol dm^{-3}. The student used a large excess of methanoic acid in the experiment. The reaction mixture was kept at a constant temperature.

iodine clock experiment 碘時鐘實驗

The student measured how the concentration of bromine varied with time and obtained results shown below.

Time (s)	Concentration of bromine (mol dm^{-3})	Rate (x 10^{-5} mol dm^{-3} s^{-1})
0	0.0100	—
50	0.0085	2.9
100	0.0072	2.4
200	0.0050	1.7
300	0.0035	1.2
400	0.0024	0.8

a) Explain why the student used a large excess of methanoic acid. AL 2007 Paper 2 Q3(a)(iii)

b) Suggest how the student could follow the progress of the reaction. ASL 2007 Paper 2 Q2(a)(i) AL 2007 Paper 2 Q3(a)(i)

c) Describe how the student could obtain the rate figures from a graph of concentration of bromine plotted against time. AL 2007 Paper 2 Q3(a)(ii)

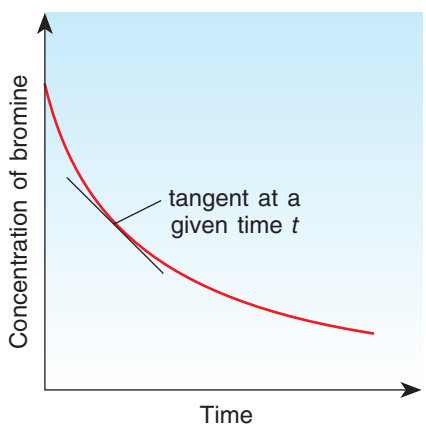
d) By plotting an appropriate graph, deduce the order of reaction with respect to bromine.

Solution

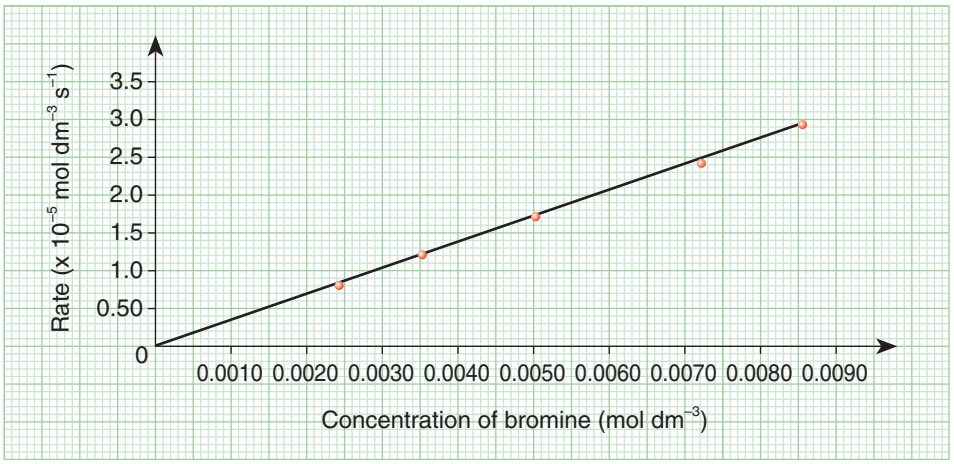
a) The student used a large excess of methanoic acid so that its concentration would not change significantly over the course of the experiment. The only variable was the change in the concentration of bromine.

b) Bromine has a yellow-brown colour. When the reaction proceeds, the yellow-brown colour of the reaction mixture becomes less intense. Concentration of bromine is directly proportional to the absorbance of the reaction mixture. Thus, we can follow the progress of the reaction using a colorimeter. DSE 2014 Paper 2 Q1(b)(ii)

c) To obtain the reaction rate at any given time, the student could draw a tangent to the concentration-time curve at that time and measure the slope of the tangent.



d) A plot of rate against concentration of bromine is shown below. It is a straight line.



Thus, the rate is directly proportional to the concentration of bromine.

i.e. rate $\propto$ $[Br_2(aq)]$

rate = $k[Br_2(aq)]$

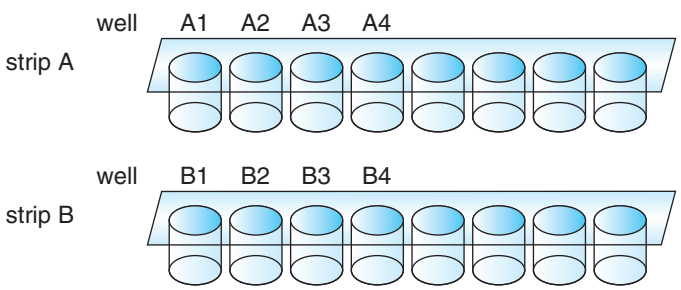
$\therefore$ the reaction is first order with respect to bromine.

45.3 Investigating the effect of changing the initial concentration of a reactant on the initial rate of a reaction

A student conducted a study on the oxidation of iodide ion by hydrogen peroxide in acidic solution.

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \longrightarrow 2H_2O(l) + I_2(aq)$$

The student performed a microscale experiment using two 8-well reaction strips, strip A and strip B as shown below.



To determine the order of reaction with respect to $H_2O_2(aq)$, the student placed chemicals in the first four wells of the two strips as shown in the table below. Then the times taken for a small, fixed amount of iodine to form upon mixing chemicals in wells bearing the same number were recorded.

Well	Number of drops			Well	Number of drops			Time (t) for a small, fixed amount of iodine to form (s)
	0.60 mol dm^{-3} $I^-(aq)$	0.080 mol dm^{-3} $S_2O_3^{2-}(aq)$	Starch solution		1.5% $H_2O_2(aq)$	$H_2O(l)$	0.60 mol dm^{-3} $H_2SO_4(aq)$	
A1	1	1	1	B1	1	7	1	115
A2	1	1	1	B2	2	6	1	58
A3	1	1	1	B3	4	4	1	29
A4	1	1	1	B4	8	0	1	14

a) Suggest how the student could determine the time for a small, fixed amount of iodine to form.

b) Explain how the initial rate of reaction can be obtained from time t.

c) Deduce the order of reaction with respect to $H_2O_2(aq)$.

d) How would you determine the order of reaction with respect to $I^-(aq)$?

AL 2005 Paper 2 Q4(a)(i),(ii)

Solution

a) The student included a small amount of sodium thiosulphate solution and starch solution in each reaction mixture. The thiosulphate ion would turn any iodine formed back to iodide ion.

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

By measuring the time taken for a dark blue colour to appear, the student knew how long it took to use up all the thiosulphate ion, and thus the time for a small, fixed amount of iodine to form.

b) As the reaction had only proceeded a small part of the way to completion, we could take initial rate $\propto \frac{1}{t}$.

c)

Well	Relative concentration of $H_2O_2(aq)$	Relative initial rate ($\frac{1}{t}$) (s^{-1})
1	1	8.7×10^{-3}
2	2	1.7×10^{-2}
3	4	3.4×10^{-2}
4	8	7.1×10^{-2}

From the above data, we can see that the relative initial rate doubles when the concentration of $H_2O_2(aq)$ is doubled. Therefore the reaction is first order with respect to $H_2O_2(aq)$.

d) Mix further series of chemicals in which the volume of $H_2O_2(aq)$, $S_2O_3^{2-}(aq)$, starch solution and $H_2SO_4(aq)$ are kept constant but the volume of $I^-(aq)$ is varied. Keep the total volume of each reaction mixture constant by adding the appropriate volume of water.

45.4 Determining the overall order of reaction

Consider the reaction between nitrogen monoxide and hydrogen:

$$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(g)$$

The following table shows some information of the reaction at 800 °C.

Experiment no.	Initial concentration of NO(g) (mol dm^{-3})	Initial concentration of $H_2(g)$ (mol dm^{-3})	Initial rate of production of $N_2(g)$ (mol dm^{-3} s^{-1})
1	1.0×10^{-3}	6.0×10^{-3}	0.50×10^{-3}
2	2.0×10^{-3}	6.0×10^{-3}	2.0×10^{-3}
3	6.0×10^{-3}	1.0×10^{-3}	3.0×10^{-3}
4	6.0×10^{-3}	2.0×10^{-3}	6.0×10^{-3}

N11

a) Deduce the orders of reaction with respect to NO(g) and $H_2(g)$.

b) Derive the rate equation for the reaction.

c) Determine the overall order of reaction.

DSE 2012 Paper 2 Q1(c)(ii),(iii)
ASL 2007 Paper 2 Q2(a)(iv)
ASL 2008 Paper 2 Q3(a)(ii)
ASL 2012 Paper 2 Q4(b)
AL 2008 Paper 2 Q2(a)(ii)
AL 2012 Paper 2 Q5(b)

Solution

a) Let x be the order of reaction with respect to NO(g) and y be the order of reaction with respect to $H_2(g)$.

The rate equation for the reaction can be expressed as:

$$\text{Rate} = k[NO(g)]^x[H_2(g)]^y$$

From experiments 1 and 2, we have

$$0.50 \times 10^{-3} = k(1.0 \times 10^{-3})^x(6.0 \times 10^{-3})^y \quad \ldots\ldots (1)$$
$$2.0 \times 10^{-3} = k(2.0 \times 10^{-3})^x(6.0 \times 10^{-3})^y \quad \ldots\ldots (2)$$

Dividing (1) by (2),

$$\frac{0.50 \times 10^{-3}}{2.0 \times 10^{-3}} = \left(\frac{1.0 \times 10^{-3}}{2.0 \times 10^{-3}}\right)^x$$

$\therefore x = 2$

N11 Initial rate of reaction with respect to NO(g)
Examination questions may ask students to deduce the initial rate of reaction with respect to NO(g) based on experiment 1.

As 2 moles of NO(g) react to produce 1 mole of $N_2(g)$, the initial rate of reaction with respect to NO(g) is twice the initial rate of reaction with respect to $N_2(g)$.

Initial rate of reaction with respect to NO(g)
$= 2 \times 0.50 \times 10^{-3}$ mol dm^{-3} s^{-1}
$= 1.0 \times 10^{-3}$ mol dm^{-3} s^{-1}

From experiments 3 and 4, we have

$3.0 \times 10^{-3} = k(6.0 \times 10^{-3})^x(1.0 \times 10^{-3})^y$ ..... (3)
$6.0 \times 10^{-3} = k(6.0 \times 10^{-3})^x(2.0 \times 10^{-3})^y$ ..... (4)

Dividing (3) by (4),

$$\frac{3.0 \times 10^{-3}}{6.0 \times 10^{-3}} = \left(\frac{1.0 \times 10^{-3}}{2.0 \times 10^{-3}}\right)^y$$

$\therefore y = 1$

So, the reaction is second order with respect to NO(g) and first order with respect to H_2(g).

Alternatively,

From experiments 1 and 2, the initial rate quadruples when the concentration of NO(g) is doubled and the concentration of H_2(g) remains unchanged.

Therefore the reaction is second order with respect to NO(g).

From experiments 3 and 4, the initial rate doubles when the concentration of H_2(g) is doubled and the concentration of NO(g) remains unchanged.

Therefore the reaction is first order with respect to H_2(g).

b) Since rate $\propto [NO(g)]^2[H_2(g)]$, the rate equation is:

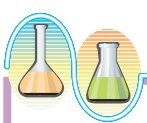
Rate $= k[NO(g)]^2[H_2(g)]$

c) The overall order of reaction is the sum of the powers in the rate equation, i.e. 2 + 1 = 3.

Notice that in Example 45.4, the order of the reaction with respect to H_2(g) is 1, even though the coefficient of H_2(g) in the balanced equation for the reaction is 2. This raises an important point:

> We cannot predict the rate equation for a reaction from its balanced equation. The only way to find the rate equation for a reaction is by doing experiments to find the order of reaction with respect to each reactant.

In the following activity, you are going to determine the rate equation for a reaction by colorimetry.



Activity 45.2

Determining the rate equation for the reaction between acidified propanone solution and iodine by colorimetry.

In the following activity, you are going to determine the rate equation for an iodine clock reaction.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 45.3

Determining the rate equation for an iodine clock reaction.

Checkpoint

Refer to page T19.

In an experiment to study the kinetics of the reaction between iodide ion and bromate ion in acidic solution:

$$6I^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3I_2(aq) + Br^-(aq) + 3H_2O(l)$$

the time t for the formation of a very small, fixed amount of iodine in each reaction mixture was measured. Results of three experiments are listed below:

Experiment no.	Initial concentration (mol dm^{-3})			t (s)
	$[I^-(aq)]$	$[BrO_3^-(aq)]$	$[H^+(aq)]$	
1	0.0020	0.0080	0.020	97
2	0.0040	0.0080	0.020	48
3	0.0020	0.0160	0.020	46

a) Describe and explain how the time t can be found experimentally. DSE 2014 Paper 2 Q1(b)(iv)(1)

b) What are the orders of reaction with respect to $I^-(aq)$ ion and $BrO_3^-(aq)$ ion respectively?
1; 1

c) A fourth experiment was conducted using the following initial concentrations:

$[I^-(aq)] = 0.0060$ mol dm^{-3}
$[BrO_3^-(aq)] = 0.012$ mol dm^{-3}
$[H^+(aq)] = 0.020$ mol dm^{-3}

Calculate the time t required for the formation of the same amount of iodine. 22 s

d) How would you determine the order of this reaction with respect to $H^+(aq)$?

The rate constant k in the rate equation is constant for a given reaction at a particular temperature. The rate equation allows us to calculate the rate constant from the rate of reaction and concentrations of reactants. The following example shows how this can be done.



45.5 Determining the rate constant for the decomposition of dinitrogen pentoxide

The decomposition of dinitrogen pentoxide in tetrachloromethane can be represented by the following equation:

$$2N_2O_5(\text{in } CCl_4) \longrightarrow 4NO_2(\text{in } CCl_4) + O_2(g)$$

a) Suggest a method that can be used to follow the progress of the decomposition and state the underlying principle of the method.

b) The results obtained for the initial rate of formation of NO_2(in CCl_4) at different initial concentrations of N_2O_5(in CCl_4) at 318 K are given below.

Experiment no.	Initial concentration of N_2O_5(in CCl_4) (mol dm^{-3})	Initial rate of formation of NO_2(in CCl_4) (mol dm^{-3} min^{-1})
1	0.10	2.9×10^{-3}
2	0.20	5.8×10^{-3}
3	0.30	8.7×10^{-3}

i) Deduce the rate equation for the decomposition.

ii) Determine the rate constant at 318 K. DSE 2012 Paper 2 Q1(c)(iii)

AL 1999 Paper 2 Q3(a)(i),(ii)
AL 2001 Paper 2 Q3(a)(i),(ii)
AL 2002 Paper 1 Q2(b)(i),(ii)

Solution

a) Monitor the pressure of the system because both N_2O_5 and NO_2 are soluble in CCl_4 while O_2 is insoluble.

b) The rate equation for the reaction can be expressed as:

$$\text{Rate} = k[N_2O_5(\text{in } CCl_4)]^x$$

From experiments 1 and 2, we have

$2.9 \times 10^{-3} = k(0.10)^x$ ..... (1)
$5.8 \times 10^{-3} = k(0.20)^x$ ..... (2)

Dividing (1) by (2),

$$\frac{2.9 \times 10^{-3}}{5.8 \times 10^{-3}} = \left(\frac{0.10}{0.20}\right)^{x}$$

$\therefore x = 1$

i.e. the rate equation for the decomposition is

Rate = $k[N_2O_5(\text{in } CCl_4)]$

ii) From experiment 1,

$[N_2O_5(\text{in } CCl_4)] = 0.10 \text{ mol dm}^{-3}$

Rate = $2.9 \times 10^{-3} \text{ mol dm}^{-3} \text{ min}^{-1}$

Substituting these values in the rate equation, we get

$2.9 \times 10^{-3} \text{ mol dm}^{-3} \text{ min}^{-1} = k(0.10 \text{ mol dm}^{-3})$

$k = 0.029 \text{ min}^{-1}$

Checkpoint

Refer to page T20.

1 A student studied the rate of a reaction with one reactant, sulphur dichloride dioxide (SO_2Cl_2).

$SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$

a) Suggest how the student could follow the progress of the reaction.

b) The results obtained for initial rates of reaction at different initial concentrations of $SO_2Cl_2(g)$ are given below.

Experiment no.	Initial concentration of $SO_2Cl_2(g)$ (mol dm^{-3})	Initial rate of reaction (mol dm^{-3} s^{-1})
1	0.20	5.8×10^{-5}
2	0.40	1.16×10^{-4}
3	0.60	1.74×10^{-4}

i) Derive the rate equation for the reaction.

ii) Calculate the rate constant.
2.9×10^{-4} s^{-1}

iii) Suppose 0.60 mole of $SO_2Cl_2(g)$ is put in a 2.0 dm^3 flask. How much $SO_2(g)$ will be formed during the first 0.50 s?
8.7×10^{-5} mol

2 Nitrogen monoxide (NO) is formed at high temperature in car engines which run on petrol. It reacts with oxygen to produce nitrogen dioxide (NO_2):

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$$

The following table shows data obtained from four experiments to investigate the effect of varying concentrations of NO(g) and $O_2(g)$ on the rate of the reaction at 20 °C.

Experiment no.	Initial concentration of NO(g) (x 10^{-2} mol dm^{-3})	Initial concentration of $O_2(g)$ (x 10^{-2} mol dm^{-3})	Initial rate of formation of $NO_2(g)$ (x 10^{-4} mol dm^{-3} s^{-1})
1	1.0	1.0	0.7
2	2.0	1.0	2.8
3	1.0	2.0	1.4
4	3.0	2.0	?

a) i) Using results for experiments 1 to 3, deduce the order of reaction with respect to NO(g) and $O_2(g)$ respectively.
2; 1

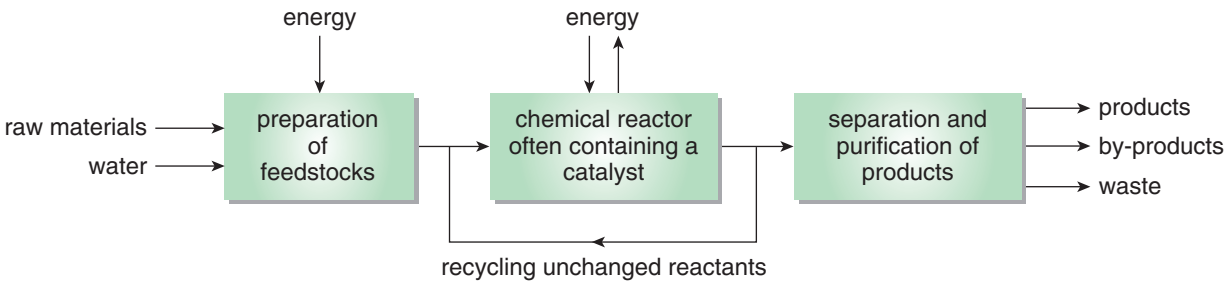
ii) Using the orders of reaction you have deduced in (i), calculate the initial rate of formation of $NO_2(g)$ in experiment 4.
1.26×10^{-3} mol dm^{-3} s^{-1}

b) Derive the rate equation for the reaction.

c) Use results for experiment 2 together with your rate equation to calculate the rate constant for the reaction.
70 dm^6 mol^{-2} s^{-1}

Summary

1 The chemical industry converts raw materials found in the Earth, ocean and atmosphere into products of greater value. The following diagram shows the stages in a chemical process.



2 A catalyst is a substance which alters the rate of reaction without itself undergoing any permanent chemical changes.

3 For a general reaction of the type:

$aA + bB \longrightarrow$ products

the rate of equation takes the form

$\text{Rate} = k[A]^x[B]^y$

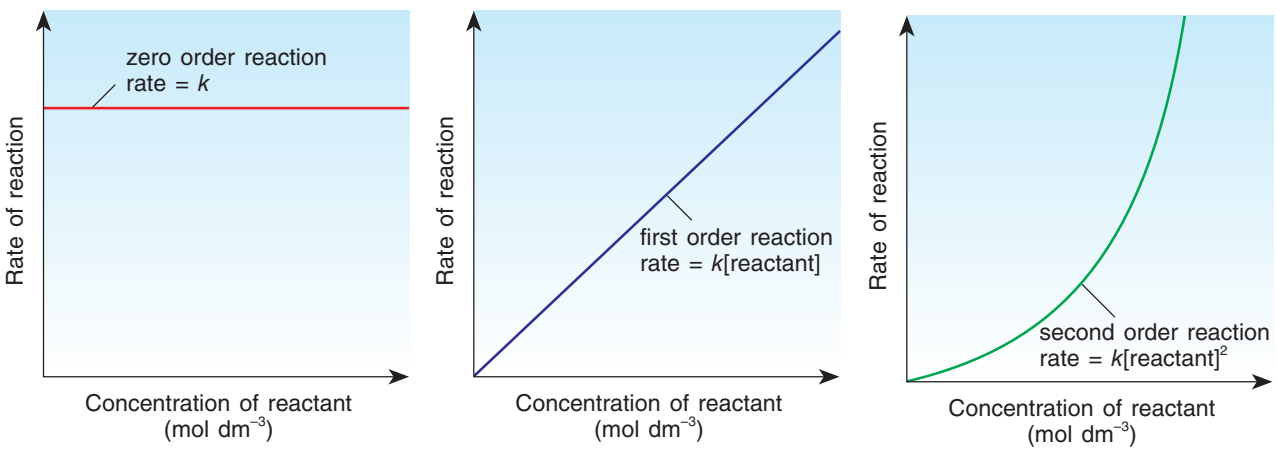
where k is the rate constant;

x is the order of reaction with respect to the reactant A;

y is the order of reaction with respect to the reactant B.

The overall order of reaction is $x + y$.

4 The following diagrams show how the rate varies with reactant concentration for zero, first and second order reactions.



5 To find the order of reaction with respect to a certain reactant, control the variables so that only the concentration of that reactant varies. All the measurements should be taken at the same temperature.

6 The initial rate of reaction is the rate of reaction at the start.

7 A way of measuring the initial rate of a reaction is to measure how long the reaction takes to produce a small, fixed amount of one of the products.

$$\text{Initial rate} \propto \frac{1}{\text{time to produce a small, fixed amount of one product}}$$

8 We cannot predict the rate equation for a reaction from its balanced equation. The only way to find the rate equation for a reaction is by doing experiments to find the order of reaction with respect to each reactant.

Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 List the FIVE categories of products from the chemical industry.

- Basic inorganic chemicals and fertilizers
- Petrochemicals and polymers
- Dyestuffs, paints and pigments
- Pharmaceuticals
- Specialities

2 List FIVE household products made from petrochemicals.

Extension Plastic household products are made from petrochemicals.

3 a) What is the difference between a batch process and a continuous process in chemical industry?
Refer to page T22.

b) Classify each of the following processes as batch or continuous.

i) The removal of carbon monoxide, oxides of nitrogen and unburnt hydrocarbons in a catalytic converter of a motor vehicle.

ii) Cake making at home.

iii) The fractional distillation of petroleum in industry to provide different fractions.

4 A chemical reaction is first order with respect to compound X and second order with respect to compound Y.

a) What is meant by the term 'order of reaction with respect to X'?
The index / power of the concentration of X in the rate equation.

b) Write the rate equation for the reaction.
Rate = $k[X][Y]^2$ where k is the rate constant.

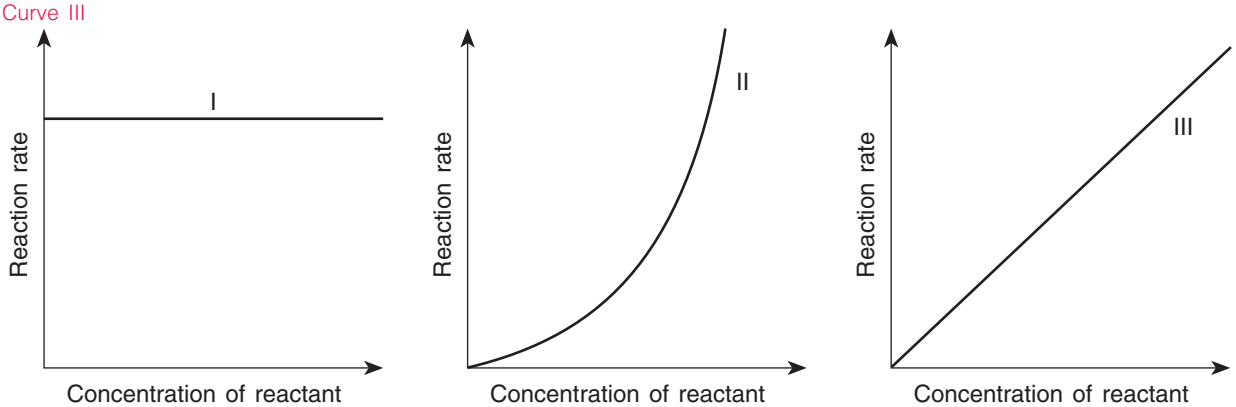
c) What is the overall order of the reaction?
3

d) By what factor will the rate increase if the concentrations of X and Y are both doubled?
8

5 Decide which curve (I, II or III) shown below corresponds to

a) a zero order reaction;
Curve I

b) a first order reaction.
Curve III



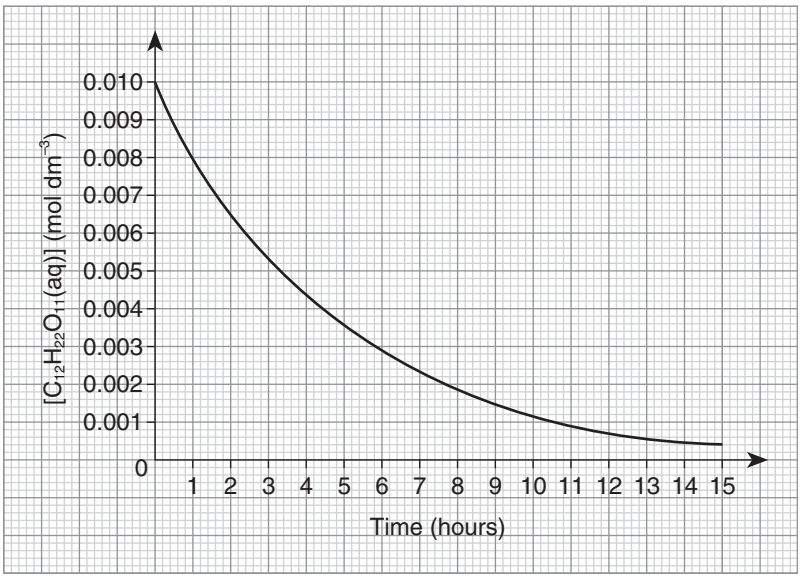
Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 In acidic aqueous solution, sucrose undergoes hydrolysis to form glucose and fructose. Refer to page T22.

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}(aq)} + H_2O(l) \longrightarrow \underset{\text{glucose}}{C_6H_{12}O_6(aq)} + \underset{\text{fructose}}{C_6H_{12}O_6(aq)}$$

An aqueous solution of sucrose, maintained at a constant acidic pH, was studied at a constant temperature. The concentration of sucrose was measured at various times. The results are shown graphically below.



a) Explain why the pH and temperature were kept constant.

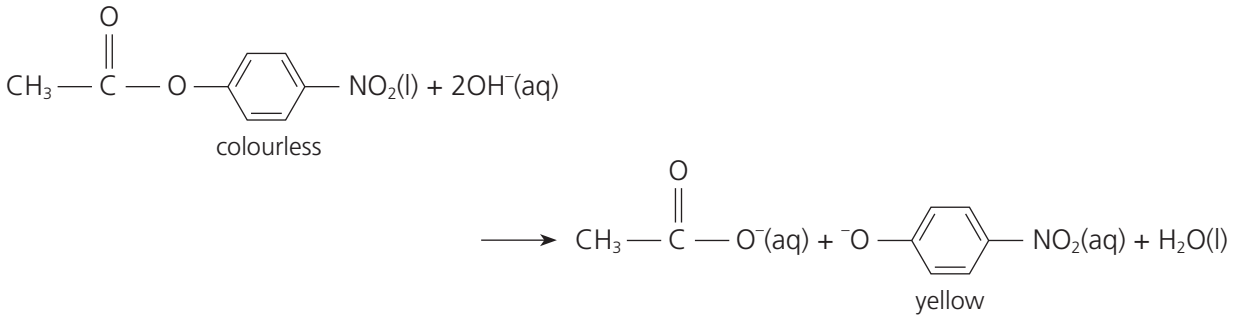
b) Find the rate of hydrolysis at FOUR different times.

c) Plot a graph of rate of hydrolysis against $[C_{12}H_{22}O_{11}(aq)]$. What order with respect to $C_{12}H_{22}O_{11}(aq)$ is suggested by this graph?
1

d) Using the graph drawn in (c), determine the rate constant.
0.21 hr^{-1}

e) Write the rate equation for the hydrolysis.

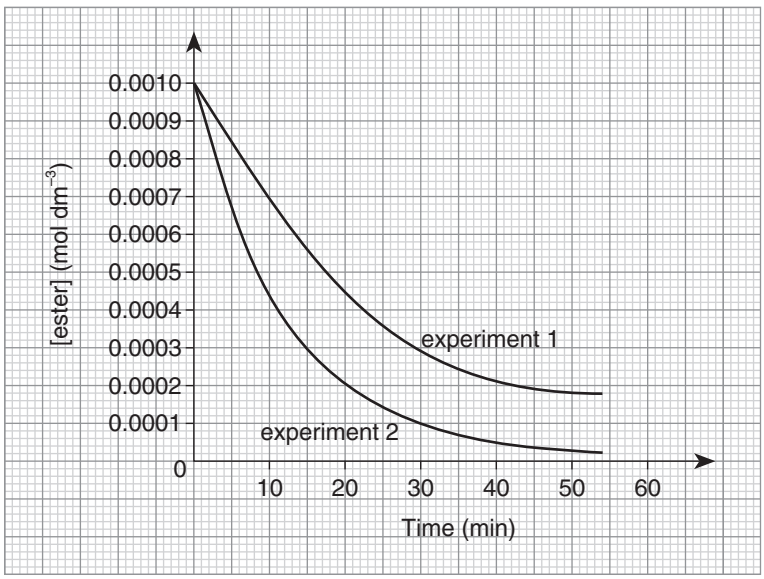
7 The ester 4-nitrophenyl ethanoate hydrolyses in alkaline solution according to the following equation: Refer to page T24.



The reaction rate was studied using two solutions of different concentrations of hydroxide ions.

Experiment 1: $[OH^-(aq)] = 0.20$ mol dm^{-3}
Experiment 2: $[OH^-(aq)] = 0.40$ mol dm^{-3}

The following graphs show how the concentration of the ester, 4-nitrophenyl ethanoate, varied over time in the two experiments.



a) Determine the initial rates of reaction during the two experiments.
3.3×10^{-5} mol dm^{-3} min^{-1}; 6.9×10^{-5} mol dm^{-3} min^{-1}

b) By using your results, deduce the order of reaction with respect to $OH^-(aq)$.
1

c) The order of reaction with respect to the ester is one. Write the rate equation for the reaction.

d) Using your rate equation and initial rates, determine the rate constant.
0.17 dm^3 mol^{-1} min^{-1}

8 The general equation for the alkaline hydrolysis of a haloalkane is as follows: Refer to page T24.

$$RX(l) + OH^-(aq) \longrightarrow ROH(aq) + X^-(aq)$$

The data below refer to a series of experiments conducted at the same temperature.

Experiment no.	Initial concentration of RX(l) (mol dm^{-3})	Initial concentration of $OH^-(aq)$ (mol dm^{-3})	Initial rate (mol dm^{-3} s^{-1})
1	0.1	0.1	1.5×10^{-6}
2	0.1	0.2	3.0×10^{-6}
3	0.2	0.1	3.0×10^{-6}
4	0.2	0.2	?

a) Determine the order of reaction with respect to the haloalkane.
1

b) Determine the order of reaction with respect to the hydroxide ion.
1

c) Determine the rate constant of the reaction.
1.5×10^{-4} dm^3 mol^{-1} s^{-1}

d) Calculate the initial rate for experiment 4.
6.0×10^{-6} mol dm^{-3} s^{-1}

e) Explain briefly why the initial reaction rate increases as each of the following increases:

i) the concentration of the haloalkane;

ii) the temperature.

9 Experiments were carried out to study the acid-catalyzed bromination of propanone at 298 K.
Refer to page T26.

$$CH_3COCH_3(aq) + Br_2(aq) \xrightarrow{H^+(aq)} CH_3COCH_2Br(aq) + HBr(aq)$$

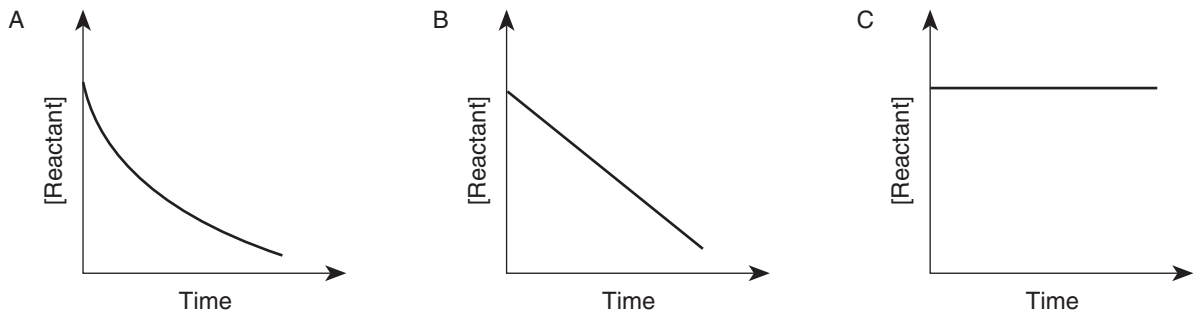
Results of the experiments are listed below.

Initial concentration (mol dm^{-3})			Initial rate (mol dm^{-3} s^{-1})
[CH_3COCH_3(aq)]	[Br_2(aq)]	[H^+(aq)]	
0.20	0.050	0.050	3.8×10^{-5}
0.20	0.10	0.050	3.8×10^{-5}
0.20	0.050	0.10	7.6×10^{-5}
0.30	0.050	0.15	1.7×10^{-4}

a) Deduce the rate equation for the reaction.

b) Calculate the rate constant of the reaction at 298 K.
3.8×10^{-3} dm^3 mol^{-1} s^{-1}

c) The following sketches show three ways in which the concentration of reagents might vary during the reaction.



Which of the above graphs correctly describes how the concentration of reactant changes with time for

i) the propanone concentration;

ii) the bromine concentration?

10 The initial rates of reaction between A and B are measured in several experiments. The following rate equation is deduced. Refer to page T27.

Rate = $k[A]^2[B]$

a) Complete the data table for the reaction.

Experiment no.	Initial concentration of A (mol dm^{-3})	Initial concentration of B (mol dm^{-3})	Initial rate of reaction (mol dm^{-3} s^{-1})
1	3.0×10^{-2}	4.0×10^{-2}	1.6×10^{-5}
2	6.0×10^{-2}	4.0×10^{-2}	6.4×10^{-5}
3	3.0×10^{-2}	1.6×10^{-1}	6.4×10^{-5}
4	1.5×10^{-2}	16.0×10^{-2}	1.6×10^{-5}

b) Using the data for experiment 1, calculate the rate constant, k.
$0.44\ dm^6\ mol^{-2}\ s^{-1}$

11 The reaction between thiosulphate ion and hydrogen ion occurs as follows: Refer to page T28.

$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow S(s) + SO_2(aq) + H_2O(l)$

Experiments were carried out to determine the order of reaction with respect to the thiosulphate ion. The time t for the formation of a small but fixed amount of sulphur from each reaction mixture was measured. Results of the experiments are as follows:

Experiment no.	Volume used (cm^3)			Time t (s)
	1.0 mol dm^{-3} HCl(aq)	$H_2O(l)$	0.040 mol dm^{-3} $Na_2S_2O_3(aq)$	
1	10.0	25.0	5.0	170
2	10.0	20.0	10.0	83
3	10.0	15.0	15.0	56
4	10.0	10.0	20.0	42
5	10.0	5.0	25.0	33
6	10.0	0.0	30.0	y

a) Describe and explain how time t could be found experimentally.

b) Explain why different volumes of water were used in different experiments.

c) i) Explain how the initial rate of reaction can be obtained from the experimental results.

ii) By plotting a suitable graph, determine the order of reaction with respect to $S_2O_3^{2-}(aq)$ ion.
1

iii) From your graph, estimate the value of y in the table.
28

Exercise

12 The reaction between iodide ion and hydrogen peroxide occurs as follows: Refer to page T30.

$2I^-(aq) + H_2O_2(aq) + 2H^+(aq) \longrightarrow I_2(aq) + 2H_2O(l)$

Initial rates for this reaction can be determined by a method known as an 'iodine clock'.

Experiments were carried out to determine the order of reaction with respect to the iodide ion. The results obtained are shown below.

Experiment no.	Volume (cm^3)						Time *t* (s)
	0.10 mol dm^{-3} KI(aq)	$H_2O(l)$	0.10 mol dm^{-3} $H_2O_2(aq)$	1.0 mol dm^{-3} $H_2SO_4(aq)$	0.010 mol dm^{-3} $Na_2S_2O_3(aq)$	Starch solution	
1	5.0	20.0	5.0	10.0	5.0	1.0	100
2	10.0	15.0	5.0	10.0	5.0	1.0	48
3	15.0	10.0	5.0	10.0	5.0	1.0	31
4	20.0	5.0	5.0	10.0	5.0	1.0	23

a) Describe and explain how time *t* could be found experimentally.

b) Explain why the volume of the reaction mixture was kept constant for all the experiments.

c) i) Explain how the initial rate of reaction could be obtained from the time *t*.

ii) By plotting a suitable graph, show that the reaction is first order with respect to $I^-(aq)$.

iii) Calculate the rate constant.
$4.9 \times 10^{-4}\ s^{-1}$

d) What further experiments will you carry out in order to determine the orders of reaction with respect to hydrogen peroxide and to hydrogen ion?

13 In an experiment to study the kinetics of the reaction:

$2I^-(aq) + S_2O_8^{2-}(aq) \longrightarrow I_2(aq) + 2SO_4^{2-}(aq)$

the time (*t*) for the formation of a very small but fixed amount of I_2 from different mixtures of KI(aq) and $K_2S_2O_8(aq)$ was measured. The results of three experiments are listed below:

Experiment no.	Initial concentration (mol dm^{-3})		*t* (s)
	$[I^-(aq)]_0$	$[S_2O_8^{2-}(aq)]_0$	
1	0.080	0.080	22.0
2	0.080	0.040	44.0
3	0.040	0.040	88.0

a) Describe and explain how time (t) can be found experimentally.

b) Deduce from the above information the rate equation for the reaction of $I^-(aq)$ with $S_2O_8^{2-}(aq)$.

c) A fourth experiment was conducted using a mixture with $[I^-(aq)]_0$ and $[S_2O_8^{2-}(aq)]_0$ of 0.100 mol dm^{-3} and 0.060 mol dm^{-3} respectively. Calculate the time (t) required for the formation of the same amount of I_2.

(HKALE 2005)

14 Consider the reaction below:

$$Br_2(aq) + HCO_2H(aq) \longrightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$$

a) Suggest an experimental method to follow the change in concentration of $Br_2(aq)$ in the reaction mixture. Given a reason for your suggestion.

The table below lists the experimental data obtained at a certain temperature:

Experiment no.	Volume used (cm^3)			Initial rate for the disappearance of $Br_2(aq)$ (mol dm^{-3} s^{-1})
	0.010 M $Br_2(aq)$	0.20 M $HCO_2H(aq)$	$H_2O(l)$	
1	2.0	10.0	8.0	1.2×10^{-5}
2	4.0	10.0	6.0	2.4×10^{-5}
3	8.0	10.0	2.0	4.8×10^{-5}

b) Suggest how the initial rate for the disappearance of $Br_2(aq)$ can be found.

c) Why is it necessary to keep the concentration of $HCO_2H(aq)$ much higher than that of $Br_2(aq)$?

d) Deduce the order of the reaction with respect to $Br_2(aq)$.

e) Suggest how the order of the reaction with respect to $HCO_2H(aq)$ can be determined.

(HKALE 2007)

15 Peroxodisulphate ion ($S_2O_8^{2-}$) slowly oxidizes arsenic(III) acid (H_3AsO_3) in aqueous solution according to the equation: Refer to page T32.

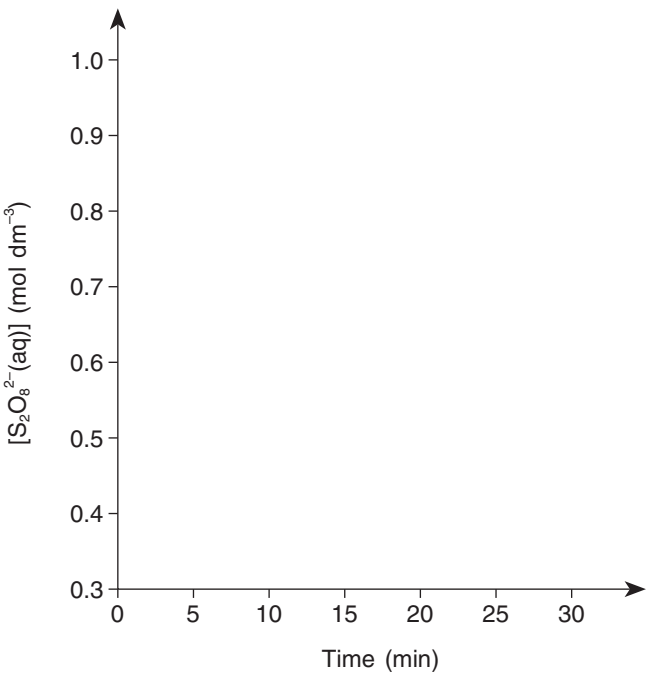
$$S_2O_8^{2-}(aq) + H_3AsO_3(aq) + H_2O(l) \longrightarrow 2SO_4^{2-}(aq) + H_3AsO_4(aq) + 2H^+(aq)$$

- 25 cm^3 of a 1.0 mol dm^{-3} solution of potassium peroxodisulphate was mixed with 25 cm^3 of a solution of arsenic(III) acid of the same concentration.
- At timed intervals, small portions of the reaction mixture were analyzed to determine the concentration of peroxodisulphate ion, $[S_2O_8^{2-}(aq)]$.

The results are shown below.

Time (minutes)	$[S_2O_8^{2-}(aq)]$ (mol dm^{-3})
0	1.0
5	0.76
10	0.62
15	0.52
20	0.44
30	0.35

a) i) On the grid below, plot a graph of the concentration of peroxodisulphate ion against time.



ii) Draw a tangent to the curve at the point where $[S_2O_8^{2-}(aq)]$ = 0.50 mol dm^{-3} and use it to calculate the slope of the curve at that point.
-0.015 mol dm^{-3} min^{-1}

b) The rate of the reaction at a particular concentration of reactant can be measured from the slope of the graph at that concentration.

The initial rate, which is the rate when the concentrations of peroxodisulphate ion ($S_2O_8^{2-}$) and arsenic(III) acid (H_3AsO_3) are both 1.0 mol dm^{-3}, was found to be 0.060 mol dm^{-3} min^{-1}.

i) Use your answer to (a)(ii) to show that the ratio of the initial rate to the rate when $[S_2O_8^{2-}(aq)]$ and $[H_3AsO_3(aq)]$ are 0.5 mol dm^{-3} is approximately 4 : 1. Use this ratio to deduce the overall order of reaction.
2

ii) Suggest TWO rate equations that agree with the order of the reaction that you have deduced in (i). How could the experiment be adapted to distinguish between these two rate equations?

(Edexcel Advanced GCE, Unit Test 6B, Jun. 2005)